

Chinese-SkillSpan: a benchmark for competency span extraction in Chinese job advertisements

Guojing Li^{†,2}, Zichuan Fu^{†2}, Wenlin Zhang², Kaifeng Guo², Xinyang Wu³, Junyi Li², and Xiangyu Zhao²

¹Renmin University of China, Beijing, China

²City University of Hong Kong, Kowloon, Hong Kong SAR, China

³Wuhan University, Wuhan, Hubei, China

Corresponding author:

Xiangyu Zhao

Email address: xianzhao@cityu.edu.hk

ABSTRACT

Competency extraction from Chinese job advertisements requires context-sensitive decisions about span boundaries and types, especially in coordinated expressions. We introduce Chinese-SkillSpan, a benchmark combining character-level source-text annotations, an explicit annotation handbook, and a shared evaluation protocol. Informed by the European Skills, Competences, Qualifications and Occupations (ESCO) classification, the handbook specifies how to preserve semantically complete spans and distinguish language, knowledge-related requirements, occupational skills, and transversal competencies in context. Human coding, disagreement analysis, and adjudication shaped these rules. The resource separates human reference annotations for evaluation from Silver labels generated by large language models for supervision. Structural checks and sampled human review assessed Silver quality, while a separate blinded study examined annotation agreement under the final handbook. Experiments compared prompted language models, Silver-supervised adaptation, and multilingual encoders. On the human reference, Silver-supervised Qwen adaptation improved typed span extraction over its unadapted baseline under the same inference settings. Blinded annotation showed greater agreement on span boundaries than on complete typed spans, indicating additional difficulty in type assignment. The human reference informed handbook development, so evaluation on new source documents remains necessary to assess generalization. Chinese-SkillSpan provides explicit annotation conventions and shared scoring procedures for diagnostic model comparison and analysis of boundary and type errors.

INTRODUCTION

Online job advertisements describe the knowledge and skills employers seek and provide evidence of changing occupational requirements (Podjanin and Strietska-Ilina, 2020; Senger et al., 2024). Document-level methods retrieve skills from predefined inventories (Bhola et al., 2020), whereas span extraction locates requirement mentions in the source text (Gnehm et al., 2022). Job Skill

[†]Guojing Li and Zichuan Fu contributed equally to this work.

Named Entity Recognition (JobSkillNER) combines span identification with type assignment. Preserving source wording allows readers to inspect span boundaries and types in context. Even short Chinese phrases can require multiple decisions about span boundaries and types. A technology name may refer to tool use in one clause and knowledge of underlying principles in another. Coordinated expressions may describe separate activities or share an action that must be retained with its objects. Chinese text generally lacks spaces between words, and tokenization alone cannot resolve these semantic boundaries (Peng and Dredze, 2015; Chen and Kong, 2021). Although similar ambiguities occur in other languages (Nguyen et al., 2024), translating an existing dataset does not by itself determine which Chinese substrings should be annotated. Explicit, context-sensitive annotation rules are therefore needed.

Existing resources provide useful foundations, but their annotation schemes and prediction targets differ. English SkillSpan permits overlapping SKILL and KNOWLEDGE annotation layers (Zhang et al., 2022b). Kompetenzer adds fine-grained labels derived from the European Skills, Competences, Qualifications and Occupations (ESCO) classification to annotated Danish and English spans through distant supervision (Zhang et al., 2022a). Work on German job advertisements extracts education, experience, and language requirements and maps their skill content to taxonomy concepts (Gnehm et al., 2022). Document-level skill retrieval and Chinese skill-demand forecasting address different prediction targets (Bhola et al., 2020; Chen et al., 2024). These differences motivate a Chinese resource with explicit conventions for span boundaries and competency types, coupled with a shared protocol for comparing extraction systems.

We introduce Chinese-SkillSpan, a benchmark for competency span extraction in Chinese job advertisements. Developed from a source corpus of 22,840 sentences across four recruitment collections, it combines annotation guidelines, separate supervision and reference subsets, scoring protocols, and baseline models (Figure 1). The handbook distinguishes language (L), knowledge-related requirements (K), occupational skills (S), and transversal competencies (T). We use *competency spans* to denote contiguous source-text mentions across all four categories and reserve *occupational skills* for category S. The annotation scheme draws on ESCO’s high-level categories and the knowledge–skill distinction in the European Qualifications Framework (European Commission, 2024; Council of the European Union, 2017), but does not assign ESCO concept identifiers. Character offsets link each annotated mention to its exact source wording. The project K category includes recruitment qualifications and explicit industry-background requirements.

The study makes three contributions: a source-text resource with separate human and Silver annotation layers, an operational handbook for Chinese boundary and type decisions, and annotation-quality evidence. The handbook specifies how to preserve semantically complete spans and assign types in context, with positive and negative examples. A separate blinded study involving three annotators assesses agreement under the final handbook, while historical coding and assisted review document human-reference construction and quality checks on Silver labels generated by large language models. Prompted models, matched Qwen adaptation experiments, and comparisons of encoders trained with Chinese supervision provide empirical support, with diagnostic analyses covering competency types, span lengths, and sentences with no reference spans.

The human reference supports model comparisons and also informed guideline development. Silver labels support training, checkpoint selection, and supplementary evaluation against generated annotations. The blinded study assesses agreement among individual annotations under the final

handbook. Table 3 summarizes the subsets and their roles, following guidance on dataset documentation and annotation agreement (Bender and Friedman, 2018; Gebru et al., 2021; Artstein and Poesio, 2008).

RELATED WORK

Skill extraction and recruitment datasets. Recruitment datasets differ in their prediction targets. Sayfullina et al. (2018) determine whether a candidate soft-skill phrase refers to an applicant, whereas Bhola et al. (2020) retrieve skills from a predefined inventory at the advertisement level. For span extraction, SkillSpan annotates English job advertisements in two potentially overlapping layers, SKILL and KNOWLEDGE. The dataset covers hard and soft skills, but these are not separate annotation labels (Zhang et al., 2022b). Kompetencer combines manually annotated skill and knowledge spans with ESCO-derived fine-grained labels in Danish and English; its distantly supervised Danish test labels were manually corrected (Zhang et al., 2022a). Gnehm et al. (2022) extract spans with coarse EDU/EXP/LNG labels and refine skill areas for taxonomy matching. Chinese recruitment research includes both forecasting and entity extraction. Job-SDF evaluates skill-demand forecasting at several aggregation levels (Chen et al., 2024). For occupational profiling, Cao et al. (2021) manually annotated skill and specialty requirements using the BIOES scheme and trained a BERT-BiLSTM-CRF model. Taxonomy-based weak supervision can reduce manual annotation effort (Zhang et al., 2022c), but the resulting labels still require validation. Nguyen et al. (2024) report difficulties with coordinated and ambiguous mentions across six recruitment datasets. Senger et al. (2024) also note the limited availability of public annotations and inconsistent terminology. Chinese-SkillSpan uses flat character-level spans with four competency types. Table 1 compares the prediction units, annotation schemes, and reported materials of selected resources.

Chinese span annotation and competency guidelines. Chinese NER resources cover social-media text (Peng and Dredze, 2015) and fine-grained entity categories (Xu et al., 2020). Multilingual studies examine entity recognition in noisy text (Fetahu et al., 2023) and changes in entity types across contexts (Luo et al., 2025). Their annotation targets differ from the L/K/S/T competency spans considered here. The absence of explicit word delimiters complicates boundary detection in Chinese (Chen and Kong, 2021); recruitment annotation also requires contextual distinctions among knowledge, occupational actions, and transversal competencies. ESCO provides a multilingual vocabulary for labor-market information (le Vrang et al., 2014; European Commission, 2024). SkiLLens combines human involvement with skill extraction and ESCO linking for European job advertisements (De Santo et al., 2026). Chinese-SkillSpan draws on ESCO’s high-level categories for Chinese L/K/S/T span annotation, without assigning ESCO concept identifiers. The handbook specifies when a tool mention should receive S rather than K, when coordinated objects must retain a shared action, and when an experience qualifier belongs within a span. Agreement analysis and adjudication address both span boundaries and type assignments, corresponding to unitization and categorization, respectively (Artstein and Poesio, 2008).

Model-assisted annotation and extraction models. Large language models (LLMs) can provide initial annotations, although their usefulness depends on task instructions and human validation (Ding et al., 2023; Li et al., 2023; Ul Haq et al., 2026). GoLLIE examines how annotation

Table 1. Selected recruitment resources: prediction units, annotation schemes, and reported materials. Table 3 reports the sizes and roles of the Chinese-SkillSpan subsets.

Study / language	Prediction unit	Annotation scheme and label source	Reported materials
Sayfullina et al. (2018); English	Candidate phrases in context	Binary labels indicating whether a soft-skill phrase refers to an applicant; not open-boundary extraction	Annotated soft-skill contexts
Gnehm et al. (2022); German	Requirement spans and finer-grained skill areas	Coarse EDU/EXP/LNG labels; finer-grained skill areas and taxonomy matching	Article and released domain-adapted models
SkillSpan (Zhang et al., 2022b); English	Source-text spans	SKILL and KNOWLEDGE in two potentially overlapping annotation layers	Dataset, code, and annotation guidance
Kompetencer (Zhang et al., 2022a); Danish / English	Spans and fine-grained labels	Manually annotated skill/knowledge spans; distant labels from the ESCO API; manually corrected Danish test labels	Dataset and code
Cao et al. (2021); Chinese	Recruitment entities	Manually annotated skill and specialty entities encoded with BIOES	Article and supporting data files
Chinese-SkillSpan; Chinese	Flat character-level spans	L/K/S/T types; frozen human reference and model-generated Silver supervision	Handbook, protocols, baseline predictions, scoring code, and independent annotation exports

Notes. EDU, EXP, and LNG denote education, experience, and language requirements, respectively. For Chinese-SkillSpan, the independent annotation exports contain the individual layers from the blinded agreement study.

guidelines affect information extraction (Sainz et al., 2024), while Kim et al. (2026) study guideline refinement for LLM annotation. JobScape generates synthetic job advertisements for skill-to-taxonomy matching (Magron et al., 2024). Chinese-SkillSpan instead adds model-generated annotations to collected recruitment texts, with documented review of sampled labels. Its annotation prompts follow the handbook version used at each stage.

Encoder-based extraction can combine pretrained models with structured decoding (Devlin et al., 2019; Lafferty et al., 2001) and continued pretraining on recruitment texts (Gururangan et al., 2020; Zhang et al., 2022b). The SkillSpan study introduced the domain-adapted JobBERT and JobSpanBERT baselines (Zhang et al., 2022b). ESCOXML-R uses taxonomy-driven multilingual pretraining (Zhang et al., 2023), and NNOSE retrieves training-data representations to support extraction, including infrequent patterns (Zhang et al., 2024). Adapting these methods to Chinese L/K/S/T extraction requires task-specific protocols because the languages, annotation layers, and data splits differ from those of the original studies.

UIE uses text-to-structure generation for information extraction (Lu et al., 2022), and UniversalNER learns from distilled supervision (Zhou et al., 2024). GLiNER performs open-type extraction with an encoder (Zaratiana et al., 2024), whereas GPT-NER uses generative extraction with self-verification (Wang et al., 2025). These four methods are discussed as related work, not evaluated baselines. Evaluation Protocol details evaluated models, supervision, and scoring procedures.

CORPUS AND ANNOTATION

Task Definition and Label Schema

Given a Chinese job-advertisement sentence x , the task is to predict non-overlapping spans (s, e, t) , where $t \in \{L, K, S, T\}$. Offsets are measured in Unicode code points: s denotes the zero-based start position, and e denotes the exclusive end position. Each annotated mention must match the contiguous substring $x[s : e]$ exactly, including its spelling and internal spaces. The annotation target is the requirement stated in the advertisement.

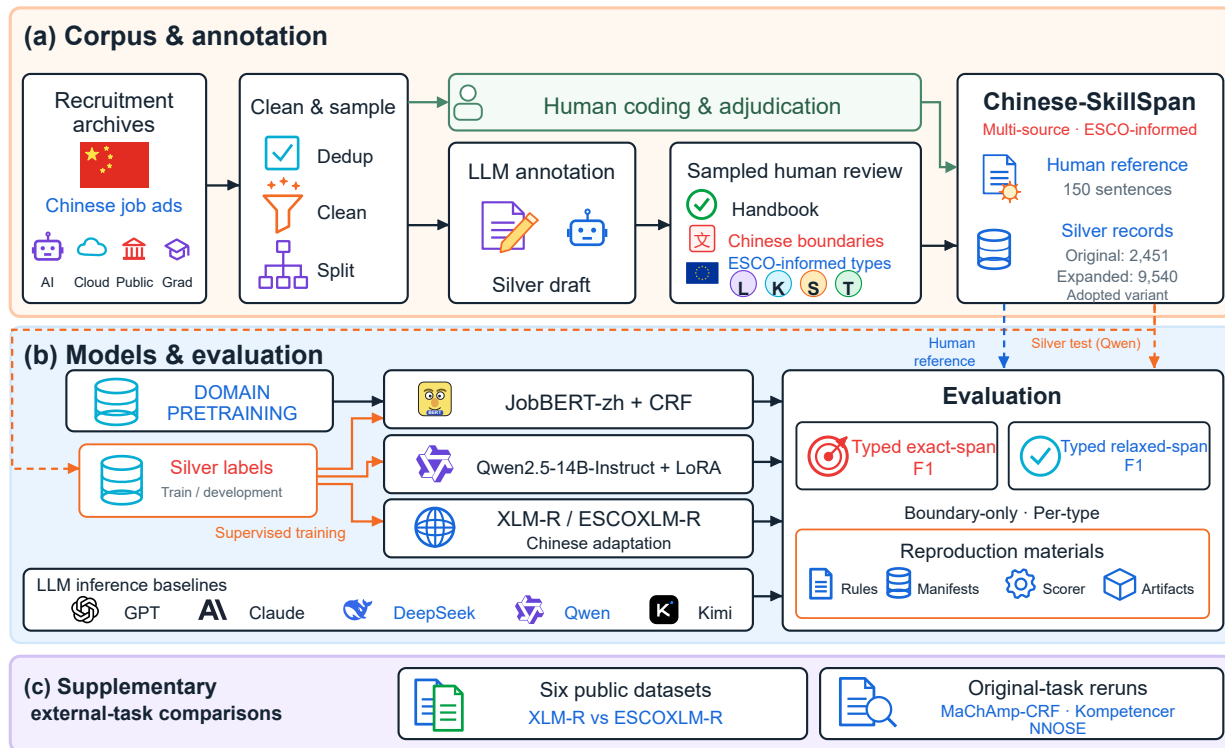


Figure 1. Overview of Chinese-SkillSpan: (a) corpus preparation and annotation; (b) model training and evaluation; and (c) external-task comparisons in the online supplement.

Figure 1 summarizes corpus preparation, annotation, and model evaluation. The four categories in Table 2 draw on ESCO and the knowledge–skill distinction in the European Qualifications Framework (European Commission, 2024; Council of the European Union, 2017).

The definitions and rule examples below illustrate the current handbook, v4.2.14, also used in the blinded agreement study. The human reference retains v4.2.10 labels, whereas the original bulk-generation prompt follows v4.2.12. Appendix A.4 gives the rules applicable to each label version. By project convention, K combines substantive knowledge with qualification and industry-background requirements. This extends the category beyond the ESCO and EQF definitions of knowledge. Scores describe the combined category; evaluating knowledge alone would require human separation of these subtypes.

Table 2. Operational L/K/S/T definitions and illustrative Chinese spans under handbook v4.2.14.

Label	Operational definition	Chinese examples and English glosses
L: Language	Natural-language names, proficiency levels, examinations, and certificates.	英语六级 (College English Test Band 6); 日语 N2 (Japanese proficiency level N2)
K: Knowledge-related requirements	Facts, principles, theories, and fields of study; educational and non-language qualifications and explicit industry-background proxies.	SQL 原理 (SQL principles); 本科及以上学历 (bachelor's degree or higher)
S: Occupational skill	Occupational activities and the application of tools or methods.	使用 [SQL] (use [SQL]); 数据分析 (data analysis)
T: Transversal competency	General communication, cooperation, cognition, self-management, and dispositions.	客户汇报 (client reporting); 责任心 (sense of responsibility)

Human annotation and LLM-assisted labeling

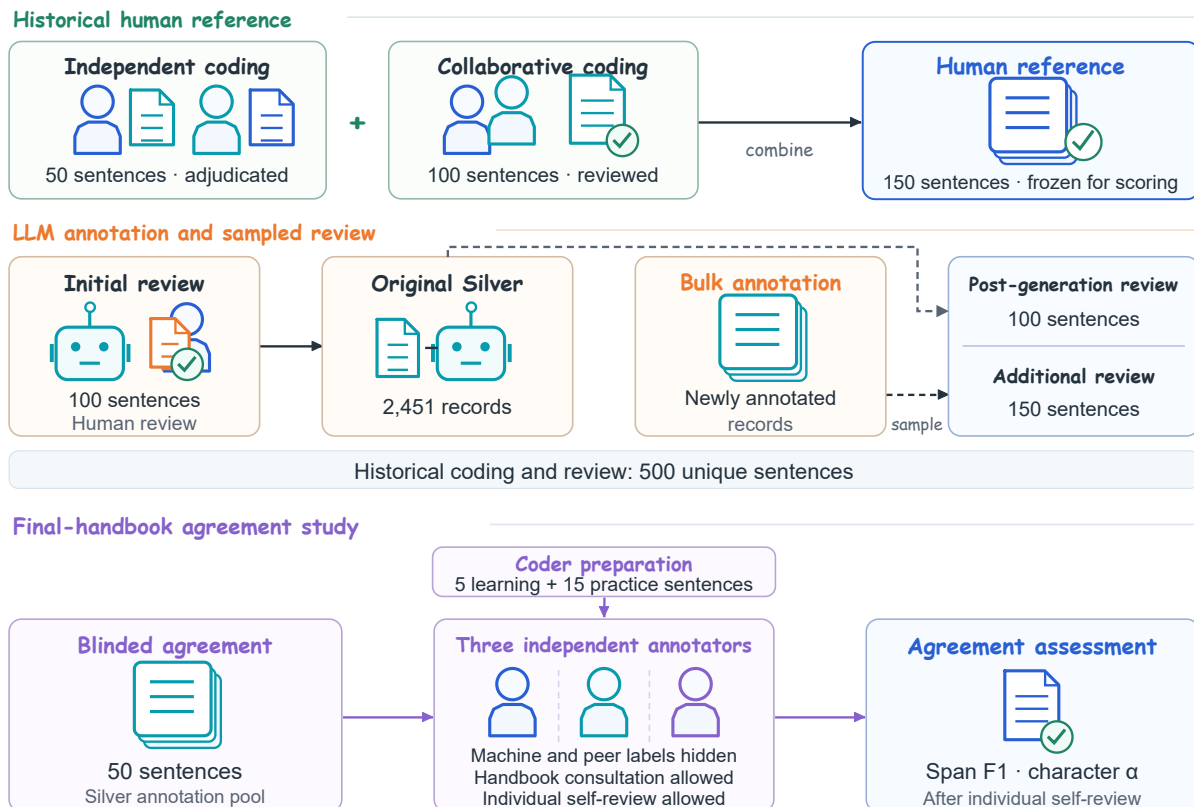


Figure 2. Human-reference construction, Silver annotation and sampled review, and the separate blinded agreement study. Vector drawing prepared with OpenAI Codex-assisted code (Use of AI Tools).

Corpus Construction and Annotation Layers

The source corpus contains 22,840 sentences from four collection batches: AI, Cloud, public-sector, and graduate recruitment. These batches can overlap in occupational coverage and acquisition route. Its original partition contains 17,460 training, 2,143 development, and 3,237 test sentences. This partition differs from the smaller labeled splits used in the experiments below; the original test partition is not used as a human-annotated test set in this study. Table 3 summarizes the corpus and its annotation, supervision, and evaluation subsets.

The texts were obtained through three acquisition routes: commercial collections purchased from MacroData (MacroData, nd); public-institution recruitment announcements collected by the research team; and the Recruitment Dataset (ID 163746) on Alibaba Cloud Tianchi (Tianchi, nd). For public-institution announcements, team members selected pages from government portals, university and hospital websites, and public recruitment platforms across regions of China. A crawler revisited the selected URLs weekly or monthly. The collected texts were screened and cleaned to form the public-institution subset. Appendix D documents the available source metadata and gaps in the collection history.

We use *original Silver* for the initial pool of generated labels and *expanded Silver* for the later training pools. The initial annotation dataset contains 2,601 records: a human reference set of 150 sentences and 2,451 records with LLM-generated Silver labels. The original Silver pool comprises 100 initially reviewed records, 730 additional records from the model-disagreement queue, and 1,621 historical extraction records. The 980-record disagreement queue supplied the human reference set and the first 830 Silver records.

The human reference set combines 50 calibration sentences with a non-overlapping 100-sentence challenge sample. The calibration sentences were independently annotated from raw text and then adjudicated; the challenge sample was collaboratively annotated and reviewed. Both samples informed guideline development. The finalized labels serve as the fixed reference for diagnostic evaluation.

The reference set contains 663 spans: 448 S, 125 K, 88 T, and two L. Eight sentences have no annotated span. All 150 complete sentence texts are distinct after NFC normalization. The initial 100-sentence review and the separate 100-sentence post-generation review do not overlap with the reference set, giving 350 unique sentences with human annotation or review at this stage. The historical 15-sentence assisted-review subset is included in this coverage and is not counted separately. Appendices B and D provide the sample counts and sampling details. Figure 2 summarizes reference construction, Silver annotation and review, and the separate agreement study.

Expanded supervision and annotation profile. The adopted expanded Silver pool contains 9,540 annotation records, combining earlier supervision with additional cloud, public-sector, and listed-company data. Candidate selection prioritized requirement-related keywords. An alternative 9,646-record pool differs in annotation channel and included records. Appendix D gives the sampling and inclusion criteria for both pools.

The additional review sample comprises 150 newly annotated sentences, stratified into 60 cloud, 45 public-sector, and 45 listed-company sentences. Checks against identifiers and normalized texts excluded the original 350 human-annotated or reviewed sentences. This sample supplements the original 100-sentence post-generation review shown in Figure 2. Historical human annotation and review cover 500 unique sentences. The separate 50-sentence blinded agreement sample has no text overlap with these cohorts, bringing verified human annotation or review coverage to

550 distinct sentences. This total counts each sentence once across annotation layers. The 150-sentence human reference set supplies the model-evaluation labels. The reproduction materials provide the cohort reconciliation.

The human reference set was selected from a model-disagreement queue, and its annotation profile differs from that of the supervision data. In the Silver training and development sets used by Qwen and the encoder comparison, 40.4% of training records and 48.5% of development records

Table 3. Data subsets and their roles. Counts can overlap across rows, and some records repeat a sentence.

Subset	Size	Composition	Use
<i>A. Corpus, human reference, and supervision</i>			
Original corpus	22,840 sentences	Four collection batches; original train/development/test partition	Source corpus
Human reference	150 sentences	50 calibration and 100 challenge sentences; 663 spans; handbook v4.2.10 labels	Primary diagnostic model evaluation
Original Silver pool	2,451 records	100 initially reviewed; 730 additional disagreement-queue records; 1,621 historical extraction records; bulk prompt v4.2.12	Candidate supervision; experiment-specific inclusion
JobBERT Silver train / dev	2,156 / 169 records	Silver supervision; historical label comparison in Appendix C	Training and checkpoint selection
Qwen / encoder Silver train / dev	2,150 / 169 records	3,646 / 345 spans; 869 / 82 records with no spans	Adapter / encoder training and checkpoint selection
Adopted expanded Silver	9,540 records	21,101 spans; 2,492 records with no spans (26.1%)	Training, checkpoint selection, and evaluation against Silver targets
<i>B. Human review and annotation agreement</i>			
Initial Silver review	100 sentences	Versioned review layers (Appendix B)	Review of generated annotations, separate from model evaluation
Original Silver post-generation review	100 sentences	144 machine-proposed spans and 148 reviewed spans	Sampled machine–human quality check
Additional review	150 sentences	Two complete reviewer submissions; 101 identical sentence-level span sets	Assisted review of newly generated annotations
Blinded agreement	50 sentences	Three annotators using handbook v4.2.14; 5 familiarization and 15 practice sentences excluded	Agreement under the final handbook; individual annotation layers

Notes. The alternative expanded Silver pool contains 9,646 records; its experimental splits appear in Appendix C.

have no annotated span, compared with 26.1% in the full expanded pool and 5.3% of reference sentences (Table 3). Occupational skills account for 54.6% of spans in the expanded pool and 67.6% in the reference set. The median span length is four Unicode code points in the reference set and six in the original Silver training/development sets. We therefore report type- and length-specific diagnostics alongside aggregate scores.

Figure 3(a,b) compares input and span lengths in the Silver training/development sets, human reference set, and additional review sample. Panel (c) includes the full adopted expanded pool in the competency-type comparison. Appendix B reports split-level counts and the two reviewers' type distributions. Repeated records retain their record weights. Lengths count Unicode code points, including punctuation and spaces, and are not directly comparable with English token counts.

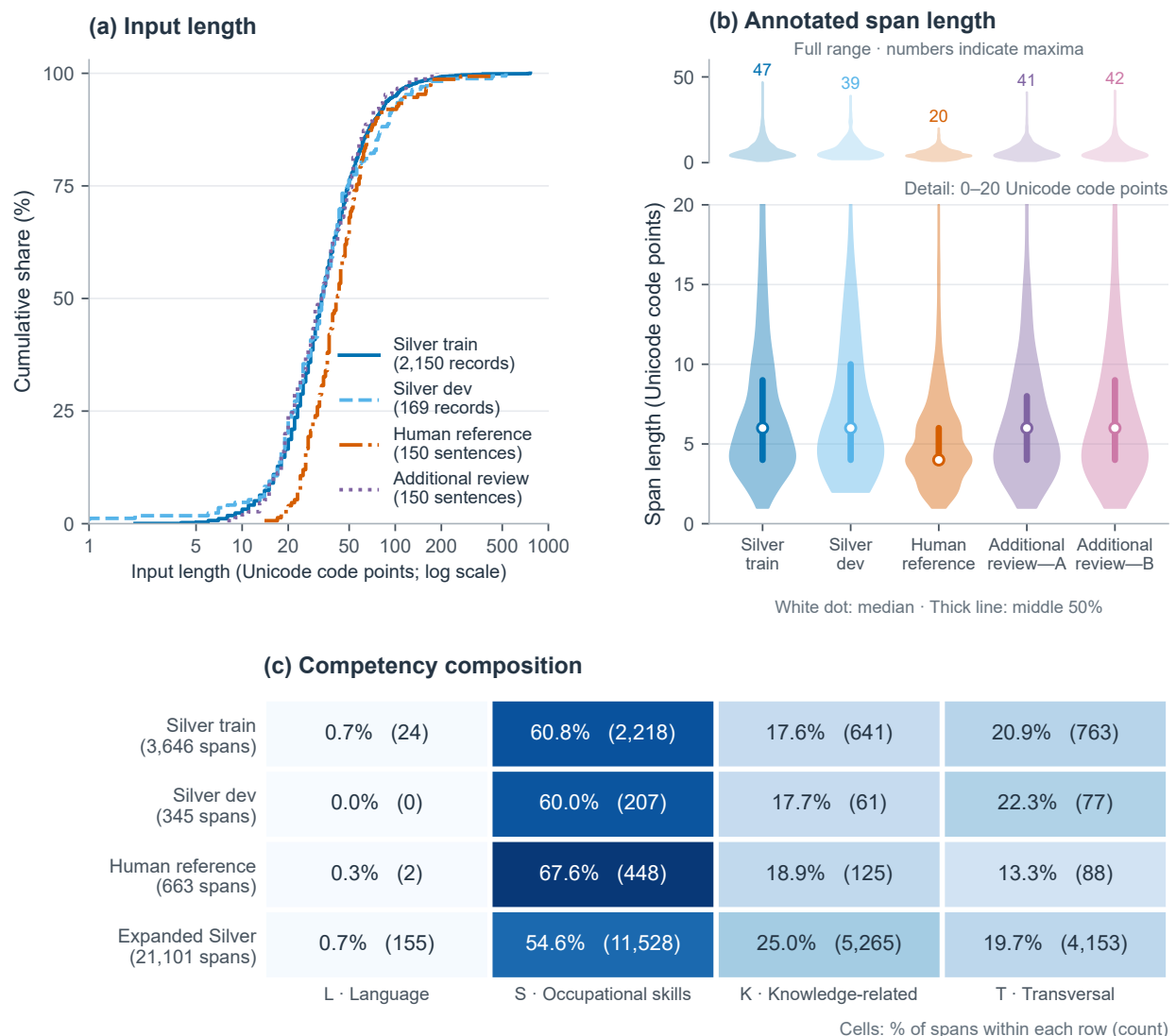


Figure 3. Annotation profiles: (a) input lengths; (b) span lengths; (c) competency-type distributions, including the full expanded pool. Silver train/dev denote the Qwen and encoder splits; A and B denote reviewers of the same sample.

Annotation Guidelines and Adjudication

Scope is determined clause by clause. Benefits, application procedures, company promotion, and outcomes alone are excluded; recruitment activities may qualify when they are occupational duties. Thus, 五险一金 (social insurance and housing fund benefits) receives an empty annotation, whereas 能力提升 (capability improvement) without sufficient context remains unresolved. Confirmed absence of an in-scope mention is distinguished from missing context or uncertain interpretation, including in mixed clauses.

Boundary selection prioritizes semantic completeness. Following SkillSpan (Zhang et al., 2022b), annotators retain the shortest complete source span, with no fixed length limit or added words.

Generic proficiency triggers usually remain outside the span, but necessary heads and complements remain inside; 会聊天 (can converse) retains its modal as a complete capability expression.

Coordinated items are split only when each remains interpretable in context. For example, 熟悉使用 Word, Excel, PPT 等办公软件 (familiar with office software such as Word, Excel, and PowerPoint) permits separate Word, Excel, and PPT spans. In contrast, 优化曝光与转化率 (optimize exposure and conversion rate) remains one S span because the objects require the shared optimization action.

Experience wording is assessed by distinguishing occupational activities, generic tenure, and industry background. Generic duration alone is excluded; explicit industry-background requirements receive the project-specific K label. For a named activity, annotators remove the experience qualifier only if the remaining words still express the requirement in context. Thus, 机器人竞赛经验 (robotics competition experience) retains its qualifier when the bare event name would change the requirement. Coordinated items must each pass the same test. Appendix A illustrates how internet-company operations receive S while an industry-background requirement receives K. Type depends on the requirement in context: 使用 SQL (use SQL) supports S for SQL, whereas 了解 SQL 原理 (understand SQL principles) supports K for SQL 原理 (SQL principles). A predicate can determine type while remaining outside the span; familiarity alone does not determine type. General communication and coordination receive T even in duty clauses, whereas technical operations receive S. Adjudication resolves scope, boundaries, and type separately, without a global label priority (Artstein and Poesio, 2008). Once scope and boundaries are resolved, type-only uncertainty may receive provisional S in the working record, with alternatives logged. Records with unresolved labels, including provisional S, are excluded in full from final training targets. Alternative annotations, nested candidates, and unresolved overlaps remain separate from the final flat layer. Figure 4 shows frozen human-reference examples; Appendix A retains the complete G1–G15 rules.

Human Annotation and LLM-Assisted Labeling

Human annotation, review, and adjudication established the human reference set and informed handbook development. The LLM-labeling route in Figure 1 concerns Silver supervision. The adopted annotation route used OpenAI Codex; Table 5 identifies the models and access routes. An initial sample review preceded bulk annotation. Automated checks examined span text, offsets, labels, and record coverage, while sampled human review assessed the interpretation of requirements. Each annotation batch used the handbook version available at the time of generation, with later amendments documented separately.

The adopted expanded Silver pool includes a replacement batch annotated through Codex. The

a) Language and transversal requirements

1802-s0008 | excerpt

4、良好的英文读写能力^L 5、优秀的沟通能力^T 和团队合作精神^T

English reading and writing are labeled L, while communication and teamwork are labeled T.

b) Knowledge and programming skills

1806-s0005 | full sentence

2 深刻理解面向对象^K、熟悉设计模式^K、多线程和跨平台编程^S；

Object orientation and design patterns are labeled K; multithreaded and cross-platform programming is retained as one S span.

c) Coordinated occupational actions

1803-s0002 | full sentence

3、独立操作竞价^S，如扩展长尾关键字^S，调整价格^S，优化曝光与转化率^S，并协助完成Listing优化^S；

Bid management, long-tail keyword expansion, price adjustment, exposure and conversion optimization, and listing optimization are all labeled S.

d) Colloquial and interpersonal capabilities

1973-s0006 | excerpt

4、会聊天^T，有服务意识^T，善于与人沟通^T，具备较好的协调^T、沟通能力^T；

Conversation, service awareness, interpersonal communication, and coordination are labeled T.

e) Experience requirements

1896-s0005 | full sentence

2、5年以上售前顾问^S或项目管理^S经验，有一定数量的大项目售前^S经验；

The sentence requests presales or project-management experience; the archived spans exclude the duration and experience qualifiers.

Figure 4. Examples of L/K/S/T span annotations from the human reference set, with English explanations. Highlights reproduce the v4.2.10 annotations used in evaluation. In (e), the final span denotes presales work on large projects.

alternative pool uses an API gateway for this batch; both pools share the earlier supervision components. Table 19 summarizes their composition.

Annotation Quality and Inter-Annotator Agreement

We assessed generated labels in two assisted-review comparisons and agreement under the final handbook in a separate blinded annotation study.

Original Silver post-generation review. A separate sample of 100 sentences was reviewed after bulk annotation, excluding the initial 100 reviewed records. Comparing the machine-prefilled

annotations with the human-reviewed layer yielded typed exact-span F1 of 0.959. Machine suggestions were visible, so this score measures agreement with assisted human review. Appendix B reports the sampling frame and supporting counts.

Additional review. On the additional 150-sentence sample, two reviewers submitted separate annotations with machine suggestions visible. Their pre-adjudication layers yielded typed exact-span F1 of 0.816 (Table 15).

Blinded annotation protocol. Three annotators used handbook v4.2.14 after familiarization on five sentences and practice on 15 sentences, following the training-and-feedback approach discussed by Klie et al. (2024). They then independently annotated the same 50-sentence sample from the existing Silver annotation candidates in separate Doccano projects, without access to machine suggestions or one another’s labels. The familiarization and practice sentences were excluded from the agreement calculation. The formal sample contained one sentence from each of 50 advertisement identifiers. It was designed for annotation agreement and overlaps with expanded-model training data (Appendix D). Source counts and sampling details appear in Appendix B.

Annotators could consult the handbook and revise their own annotations during individual self-review. Agreement was calculated from their separate exports after self-review, using all 50 sentences and all three annotator pairs. The study provides three individual annotation layers without an adjudicated consensus reference.

Table 4. Inter-annotator agreement on the 50-sentence blinded sample after individual self-review under handbook v4.2.14. Brackets show 95% bootstrap confidence intervals.

<i>A. Span-level agreement</i>			
Comparison	Typed exact-span F1	Boundary-only exact-span F1	Identical typed-span sets
A–B	0.793 [0.694, 0.884]	0.828	36/50
A–C	0.848 [0.752, 0.926]	0.876	39/50
B–C	0.814 [0.721, 0.899]	0.847	37/50
Mean pairwise	0.818 [0.732, 0.894]	0.850 [0.766, 0.923]	—
<i>B. Character-level agreement (2,111 Unicode code points)</i>			
Comparison	Coefficient	Value	
A–B	Cohen’s κ	0.880	
A–C	Cohen’s κ	0.930	
B–C	Cohen’s κ	0.909	
All three annotators	Nominal Krippendorff’s α	0.907 [0.851, 0.954]	

Notes. Familiarization (5 sentences) and practice (15) are excluded. Mean pairwise F1 is the arithmetic mean of the three pairwise scores. Identical sets include empty sets.

Agreement measures. We computed typed exact-span F1 for each annotator pair over the full sample and report the mean of the three pairwise scores. A match requires identical boundaries and types, without treating any annotator as the reference (Hripcsak and Rothschild, 2005). Boundary-only exact-span F1 ignores type. Character-level agreement was measured using pairwise Cohen's κ and three-annotator nominal Krippendorff's α , based on L/K/S/T/O assignments to all 2,111 source Unicode code points; O denotes positions outside annotated spans. Reported 95% percentile intervals were obtained from 10,000 source-stratified sentence-bootstrap samples, retaining all three annotation layers together for each sampled sentence. The intervals summarize variation within the sampled source strata for these three annotators.

Agreement results. Mean pairwise typed exact-span F1 was 0.818 (95% CI [0.732, 0.894]), compared with 0.850 for boundary-only exact-span F1 (95% CI [0.766, 0.923]). The higher boundary-only score indicates additional disagreement over type assignment. Pairwise typed exact-span F1 ranged from 0.762 to 0.805 for K, 0.778 to 0.869 for S, and 0.865 to 1.000 for T. Character-level α was 0.907 (95% CI [0.851, 0.954]). Character-level coefficients summarize position-wise label agreement, whereas span-level F1 requires matching complete spans (Artstein and Poesio, 2008).

All three annotators produced identical complete typed-span sets for 34 sentences, including 11 with no annotated spans. They also identified the same two L spans, both in one sentence, providing limited evidence about agreement on this category. The reproduction materials include the individual annotation exports used in these calculations and the associated diagnostics.

Silver labels against blinded annotations. In a post hoc comparison, the adopted expanded pool retained Silver labels for 49 of the 50 blinded-study sentences. Comparing these labels separately with the three blinded annotation layers yielded a mean typed exact-span micro-F1 of 0.709. This measures agreement between historical Silver labels and final-handbook annotations on these sampled texts. Appendix B reports the per-annotator results and the alternative pool.

Use of AI Tools

OpenAI Codex assisted with manuscript revision, LaTeX formatting, and code preparation and checking under author direction. Anthropic Claude helped check sentence-level wording. xAI Grok 4.6 with high reasoning effort in Cursor assisted with code preparation and checking. Figures 1 and 2 are editable vector drawings prepared with OpenAI Codex-assisted code; OpenAI's image-generation tool assisted the initial layout of Figure 1. The annotation workflow used GPT-6 Astra through OpenAI Codex to generate Silver labels. GPT-6 Astra and Grok, together with Qwen, DeepSeek, and Kimi, also provided suggestions during assisted annotation. Table 5 separates these roles from the evaluated model configurations. Available model identifiers and settings accompany the reproduction materials; some preparation-tool versions were not retained. The authors reviewed the revisions and take responsibility for the manuscript.

Ethics and Data Use

The study analyzed recruitment advertisements and did not recruit job applicants or collect applicant responses. Annotation and review were conducted by members of the research team. Institutional ethics approval was not sought, and no formal exemption is claimed. The authors confirm permission to use the source materials for academic research. Research-use permissions and

rights to redistribute third-party recruitment texts are documented separately in the access terms for each component. Appendix D reports the automated privacy-screening results and the human review still required before a privacy-vetted release.

EVALUATION PROTOCOL

The evaluation covers prompted extraction, silver-supervised Qwen adaptation, and encoder initialization for Chinese competency extraction, with diagnostics by type, span length, and false positives on sentences without reference spans. The human reference set provides primary evaluation targets; supplementary silver validation and test scores measure agreement with generated annotations.

Datasets and Splits

The original corpus was partitioned by source document into training, development, and test sets with 1,600, 200, and 200 document identifiers, respectively. These partitions share no document identifiers and differ from the later labeled splits used for Silver supervision and human-reference evaluation.

We checked the Silver training and development sets for sentence-level overlap. JobBERT-zh + CRF uses Silver sets of 2,156 training and 169 development records. For Qwen and the encoder comparison, six training records matching three development texts after NFC normalization were removed, leaving 2,150 training and 169 development records. In this split, training and development share no identical or NFC-normalized complete texts, and neither contains a human-reference sentence by identifier or normalized text.

An audit of source-document identifiers found that the 2,150-record Silver training set shared one group with development and 59 groups with the human reference set. The latter contain 143 of the 150 reference sentences. These are different sentences from shared documents, which may contain multiple job postings. Reference sentences are excluded from supervised inputs, but their source documents overlap with training data.

Supplementary expanded-pool Qwen experiments use 8,842/348/350 training/validation/test records from the adopted expanded pool and 8,943/351/352 from the alternative pool. Both combine earlier supervision with additional annotations and differ in annotation channel and included records. Records shared across annotation-channel variants keep their split assignments. Checkpoints are selected by typed exact-span F1 on Silver validation labels. Silver-test scores measure agreement with generated labels in a separate sentence split whose source-document groups overlap with training. Sentence-level checks also excluded human-reference sentences from expanded-model fitting and checkpoint selection. Expanded-model settings and results appear in Appendix C; duplicate records, normalization checks, and source-group overlaps appear in Appendix D.

Evaluation Metrics

The primary metric is typed exact-span micro-F1: a match requires identical span boundaries and L/K/S/T type. Typed relaxed-span micro-F1 uses one-to-one greedy matching of same-type spans with intersection-over-union (IoU) of at least 0.5:

$$\text{IoU}(u, v) = \frac{|u \cap v|}{|u \cup v|}, \quad \text{IoU}(u, v) \geq 0.5. \quad (1)$$

Here, u and v are the sets of Unicode-code-point positions in the predicted and reference spans; $|\cdot|$ counts positions. Under either matching rule, true positives (TP) are matched prediction–reference pairs, false positives (FP) are unmatched predictions, and false negatives (FN) are unmatched reference spans. Counts are summed across sentences before computing the micro-averaged metrics:

$$P_{\text{micro}} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad R_{\text{micro}} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \quad F_{1,\text{micro}} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}. \quad (2)$$

When a denominator is zero, the corresponding precision, recall, or F1 is set to zero. All 150 sentences in the human reference set are scored. Expanded Qwen runs also report silver validation and test scores. Accepted spans from partially rejected responses remain in scoring; malformed responses and responses with no accepted spans yield empty predictions. A sentence with no predicted or reference spans contributes no TP, FP, or FN counts. Appendix C gives parser rules and matching order and links the scoring implementation.

Boundary-only exact-span micro-F1 ignores type and matches span boundaries. Per-type scores are computed over all reference sentences, including those without any reference span of the evaluated type. Macro-F1 is the arithmetic mean of F1 over the specified categories.

For experiments repeated across three training seeds, we report the mean and sample standard deviation, where f_i is the score from run i :

$$\bar{f} = \frac{1}{n} \sum_{i=1}^n f_i, \quad s_f = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (f_i - \bar{f})^2}, \quad n = 3. \quad (3)$$

Inference-only configurations and expanded-pool experiments are single runs unless otherwise stated. For paired model comparisons, 95% percentile intervals are obtained from 10,000 bootstrap resamples of the 61 source-document groups in the human reference set. Complete groups are sampled with replacement, using the same draw for all compared models. These intervals describe uncertainty from resampling the reference set for the evaluated runs; training-seed standard deviations describe variation across training runs. Appendix C provides the full resampling procedure.

Models and Baselines

JobBERT-zh is a sequence-labeling baseline with a Chinese recruitment-domain-pretrained encoder and a CRF head. We compare Qwen2.5-14B-Instruct (Qwen) (Qwen Team, 2025) with and without a project-trained LoRA adapter (Hu et al., 2022) for generative extraction.

The prompted comparison uses shared annotation instructions across GPT-5.6 Terra, Claude Sonnet 4.5, DeepSeek-V4-Pro (DeepSeek), Kimi K2.6 (Kimi), and Qwen without a project adapter. DeepSeek is evaluated with thinking enabled and disabled, giving six configurations. GPT-5.6 Terra and Claude Sonnet 4.5 were accessed through an API gateway. Table 5 lists model identifiers, access routes, and key settings.

The adopted Silver annotation route used the Codex configuration in Table 5, and the GPT baseline used a separate API configuration. Shared guidelines and assisted review can nevertheless align annotation choices across these configurations, limiting the independence of their errors.

Table 5. Model names, recorded identifiers, and access routes by study role.

Model / role	Recorded identifier / selection	Access and key settings
<i>Evaluation configurations</i>		
GPT-5.6 Terra	Request: gpt-5.4; returned: gpt-5.6-terra	API gateway; temperature 0; cap 2,048 tokens.
Claude Sonnet 4.5	Request alias: claude-sonnet-4-5	API gateway; temperature 0; cap 2,048.
DeepSeek-V4-Pro	deepseek-v4-pro	DeepSeek API; non-thinking: temperature 0, cap 2,048; thinking: cap 8,192, temperature unspecified.
Kimi K2.6	kimi-k2.6	Moonshot API; non-thinking; temperature 0.6; cap 2,048.
Qwen (unadapted / LoRA)	Qwen2.5-14B-Instruct	Local weights; deterministic decoding; cap 2,048.
JobBERT-zh + CRF	Released initialization and Silver-trained continuation	Local encoder and inherited CRF.
<i>Annotation and assisted review</i>		
Original Silver annotation	GPT-6 Astra (gpt-6-astra)	OpenAI Codex; adopted original Silver labels.
Expansion comparison	GPT-6 Astra (gpt-6-astra)	Codex / API gateway; 2,500-candidate replacement batch; earlier components shared.
Additional review	GPT-6 Astra; Grok 4.6	Codex / Cursor; Grok: high reasoning effort; suggestions visible to reviewers.
Annotation support	Qwen; DeepSeek; Kimi	Machine-generated annotation suggestions; versions incompletely recorded.

Caps are output-token limits. Codex and Cursor denote interfaces. Gateway model names follow recorded response identifiers; upstream identities were not independently verified. Appendix C gives run settings; Appendix D records checkpoint provenance and implementation gaps.

Implementation Details

Supervision and model selection. The JobBERT-zh + CRF baseline starts from the released encoder and inherited CRF and is trained on 2,156 Silver training records, with 169 Silver development records for checkpoint selection. Its supervision incorporates updated guidelines, LLM annotations, and adjudication. Checkpoints are selected by development typed exact-span F1. Appendix C preserves the historical comparison of earlier and revised supervision.

Qwen starts from pretrained Instruct weights, with a fresh LoRA adapter for each of seeds 42, 43, and 44 on the Silver sets of 2,150 training and 169 development records. Training uses a fixed system instruction, user template, and five schematic examples. Targets contain literal span text, occurrence index, and type. Loss covers the assistant’s JSON output and closing turn token. Training uses BF16 precision; development typed exact-span F1 determines checkpoint selection. Table 6 summarizes the training settings.

Table 6. Training settings for the Silver-supervised JobBERT-zh and Qwen experiments.

Setting	JobBERT-zh + CRF	Qwen + LoRA
Learning rate	2×10^{-5}	10^{-4}
Batch size	16	1; gradient accumulation over 16 steps
Maximum input length	256 tokens	8,192 tokens
Maximum training epochs	6	6
Patience	2	2 epochs

Qwen LoRA uses rank 16, scaling 32, and dropout 0.05 on the attention q/k/v/o projections.

Inference. Each generative request contains one sentence’s identifier and text, shared instructions based on handbook v4.2.12, and five fixed examples. A parser maps occurrence-based outputs to source Unicode-code-point spans. Adapted and unadapted Qwen share the base checkpoint, prompts, parser, BF16 precision, deterministic decoding, and a 2,048-token output limit. DeepSeek limits output to 2,048 tokens with thinking disabled and 8,192 with thinking enabled. The prompted configurations use the same instructions with model-specific output budgets. Appendix C lists service settings and output checks.

Encoder Comparison with Chinese Supervision

To assess taxonomy-informed initialization, we compare XLM-R-large (Conneau et al., 2020) and ESCOXLM-R (Zhang et al., 2023) with newly initialized nine-label BIO heads on the same Silver training/development split of 2,150 training and 169 development records. Both use AdamW, a learning rate of 2×10^{-5} , batch size 8, at most 20 epochs, and seeds 42, 43, and 44. Development typed exact-span F1 determines checkpoint selection.

Both models assign BIO labels to the first subword of each original Chinese character. Predictions from chunks of at most 128 characters, including the final remainder, are combined to cover complete sentences. Tokenizer placeholders preserve whitespace and other positions that would otherwise be discarded; scoring uses original text and offsets. Human-reference sentences are excluded from fitting and checkpoint selection.

We report typed exact-span and typed relaxed-span micro-F1, boundary-only exact-span micro-F1, and macro-F1 over both L/K/S/T and K/S/T. The K/S/T sensitivity average excludes the sparsely represented L category (two reference spans); all models still predict four types. Prediction heads and training budgets differ from those of JobBERT and Qwen, so cross-architecture scores are descriptive rather than controlled comparisons. Paired encoder comparisons use the source-group bootstrap defined above.

RESULTS

Prompted Inference on the Human Reference Set

GPT-5.6 Terra achieved the highest observed typed exact-span micro-F1 among the six configurations, scoring 0.667 compared with 0.647 for DeepSeek with thinking disabled (Table 7). Their paired difference was 0.020, with a 95% source-group bootstrap interval of $[-0.027, 0.061]$ (Table 8). The interval includes zero, so the observed lead is not clearly resolved by this comparison. These end-to-end scores apply to the prompts, output budgets, and parsing rules specified in the evaluation protocol. Qwen without an adapter had the lowest recall at 0.281.

Table 7. Prompted extraction on the human reference set (single runs).

Recorded configuration	P	R	Exact F1	Relaxed F1
GPT-5.6 Terra	0.676	0.658	0.667	0.766
DeepSeek-V4-Pro (non-thinking)	0.651	0.643	0.647	0.761
Claude Sonnet 4.5	0.598	0.597	0.598	0.709
DeepSeek-V4-Pro (thinking)	0.713	0.505	0.591	0.671
Kimi K2.6	0.581	0.563	0.572	0.714
Qwen	0.507	0.281	0.361	0.470

P/R: typed exact precision/recall. Exact and relaxed F1 are typed and micro-averaged. Qwen uses no adapter. Model identifiers and settings appear in Table 5.

Table 8. Paired differences in typed exact-span micro-F1.

Comparison	ΔF_1	95% CI
GPT-5.6 Terra – DeepSeek-V4-Pro (non-thinking)	+0.020	[−0.027,0.061]
Qwen LoRA (seed 42) – Qwen (no adapter)	+0.138	[0.069,0.200]
Qwen LoRA (seed 43) – Qwen (no adapter)	+0.202	[0.133,0.269]
Qwen LoRA (seed 44) – Qwen (no adapter)	+0.197	[0.123,0.268]
Qwen LoRA (seed mean) – Qwen (no adapter)	+0.179	[0.111,0.242]
ESCOXLM-R – XLM-R-large (seed means)	+0.016	[−0.005,0.035]

ΔF_1 : first model minus second. CI: pointwise paired source-group bootstrap confidence interval (61 groups).

444 Fine-Tuning with Silver Labels

445 All three Qwen adapters improved typed exact-span micro-F1 over the unadapted baseline under
 446 the same inference protocol (Table 8; Table 9, panel A). The adapted mean was 0.540 ± 0.035 ,
 447 compared with 0.361 without an adapter. The paired seed-mean difference was 0.179, with a 95%
 448 interval of [0.111,0.242]; the three seed-specific intervals were also above zero. Adaptation also
 449 increased the mean scores under relaxed and boundary-only matching.

450 The adapters also produced fewer responses containing rejected span candidates than the un-
 451 adapted model. These end-to-end scores jointly reflect output validity and span extraction. De-
 452 tailed parser outcomes accompany the reproduction materials.

Table 9. Human-reference results for Qwen adaptation and the Silver-trained JobBERT baseline.

A. Qwen: fixed inference protocol				
Condition	Train / dev records	Typed exact	Typed relaxed	Boundary-only exact
No adapter	—	0.361	0.470	0.414
Qwen + LoRA	2,150 / 169	0.540 ± 0.035	0.662 ± 0.033	0.571 ± 0.038
B. Silver-trained JobBERT baseline				
Model	Train / dev records		Typed exact	
JobBERT-zh + CRF	2,156 / 169		0.554 ± 0.005	

F1 scores (0–1) are mean $\pm$ sample SD over seeds 42, 43, and 44; Qwen without an adapter is a single run. Boundary-only exact F1 ignores types. Per-seed Qwen scores accompany the reproduction materials.

The Silver-trained JobBERT-zh + CRF baseline achieved mean typed exact-span micro-F1 of 0.554 ± 0.005 (Table 9, panel B). Its observed mean exceeds Qwen + LoRA (0.540), while both are below prompted GPT-5.6 Terra (0.667). Their training and inference protocols differ, so these cross-architecture comparisons are descriptive.

Supplementary Analyses

Each Qwen adapter increased exact-match recall in all three reference-span length bins: ≤ 4 , 5–8, and ≥ 9 Unicode code points (Figure 5). These bins contain 346, 243, and 74 reference spans, respectively.

On the eight sentences with no reference spans, the number of false-positive spans decreased from 19 without an adapter to 6, 4, and 0 for seeds 42, 43, and 44. The unadapted run produced false positives in two sentences; the adapted runs affected two, one, and zero sentences, respectively. For seed 42, the reduction occurred within the same two affected sentences.

Encoder and Competency-Type Comparisons

XLM-R-large and ESCOXML-R achieved mean typed exact-span micro-F1 scores of 0.522 and 0.538, respectively (Table 10, panel A). Both models are evaluated against the same human reference set. The mean difference was 0.016, with a paired 95% interval of $[-0.005, 0.035]$ (Table 8). The interval includes zero, leaving the advantage of ESCO initialization unresolved under this protocol.

Four-type macro-F1 was 0.607 for XLM-R-large and 0.622 for ESCOXML-R. Averaging over K/S/T gave 0.542 and 0.562, respectively. These sensitivity summaries exclude the sparsely represented L category from the macro average without changing the four-type prediction task.

Qwen adaptation improved all three K/S/T categories (Table 10, panel B; Figure 5). The largest observed increase among these categories was for T, from 0.380 to 0.638. Mean S and K scores increased by 0.148 and 0.210, respectively. Among the three Silver-trained configurations in Table 10, panel B, Qwen had the highest observed mean scores for S and T, while ESCOXML-R had the highest K score. These rankings are descriptive because prediction heads, training budgets, and inference procedures differ across architectures.

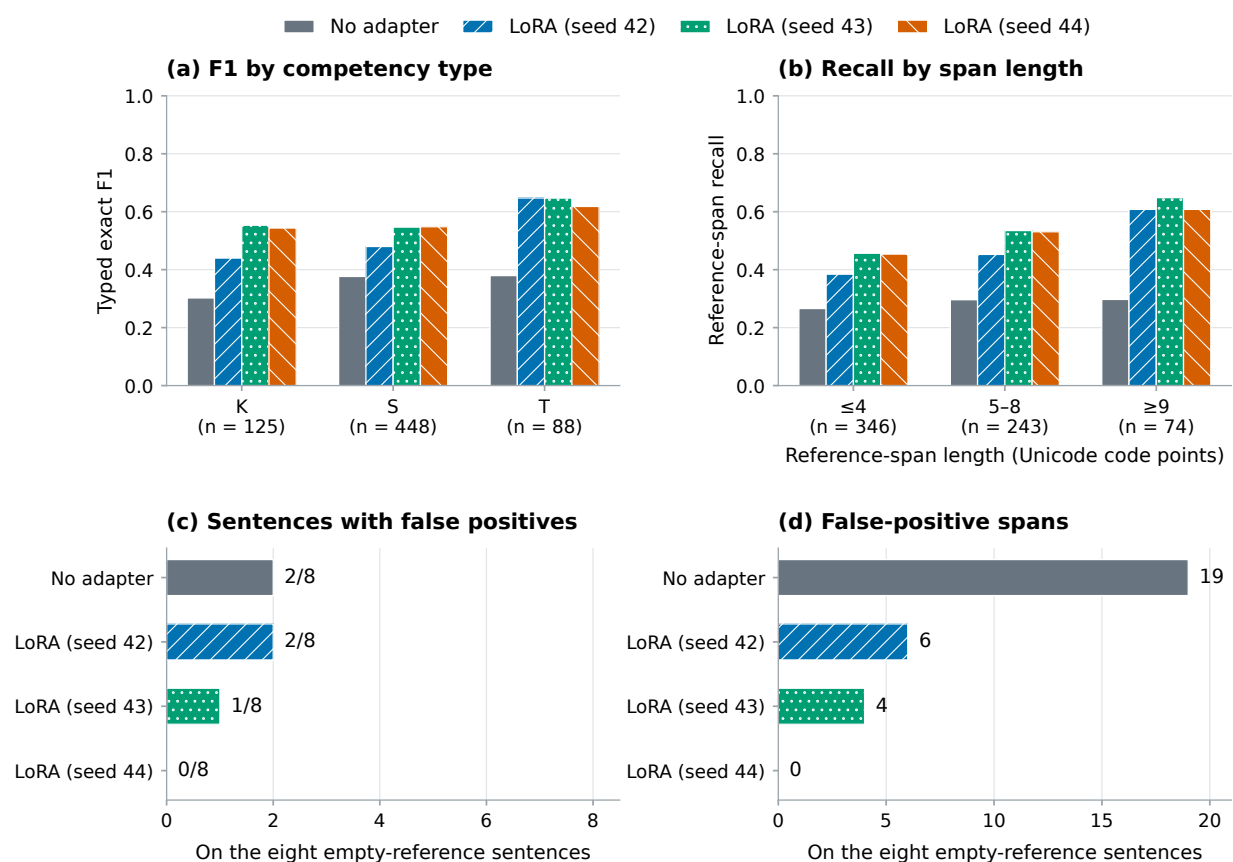


Figure 5. Qwen diagnostics on the human reference set: (a) typed exact-span F1 for K/S/T; (b) recall by reference-span length (n : reference spans); (c–d) false positives on eight sentences with no reference spans, counted by sentence and span.

Table 10. Overall and per-type results on the human reference set.

A. Encoders with Chinese supervision: overall F1					
Encoder	Typed exact	Typed relaxed	Boundary-only exact	Macro L/K/S/T	Macro K/S/T
XLM-R-large	0.522 ± 0.022	0.650 ± 0.021	0.571 ± 0.022	0.607 ± 0.016	0.542 ± 0.021
ESCOXLM-R	0.538 ± 0.021	0.662 ± 0.022	0.582 ± 0.024	0.622 ± 0.010	0.562 ± 0.013
B. Typed exact-span F1 by competency type					
Model	S ($n = 448$)	K ($n = 125$)	T ($n = 88$)	L ($n = 2$)	
Qwen, no adapter	0.377	0.302	0.380	0.000	
Qwen + LoRA	0.525 ± 0.039	0.512 ± 0.063	0.638 ± 0.017	0.800 ± 0.000	
XLM-R-large	0.506 ± 0.024	0.495 ± 0.037	0.627 ± 0.006	0.800 ± 0.000	
ESCOXLM-R	0.518 ± 0.029	0.537 ± 0.009	0.631 ± 0.020	0.800 ± 0.000	

F1 scores (0–1) are mean $\pm$ sample SD across three training seeds; unadapted Qwen is a single run. n counts reference spans. All trained runs shown have identical L counts (TP/FP/FN = 2/1/0), based on only two reference spans.

Boundary and Type Disagreements in Human Annotation

The blinded annotation study found disagreement in both span boundaries and types. Mean boundary-only exact-span F1 was 0.850, compared with 0.818 when matching also required identical types (Table 4). The decrease when type matching is required shows that type assignment contributes to disagreement beyond span boundaries.

The agreement study and model evaluation use different texts, label versions, and annotation procedures. Direct comparison of human and model agreement would require evaluating both on a common independently annotated sample.

Scoring Criteria and Evaluation Targets

For the same unadapted Qwen predictions, relaxing the matching criterion raises F1 from 0.361 to 0.470 (Table 7). Under typed exact matching, the 181 false positives comprise 27 same-boundary/different-type pairs, 89 same-type/overlapping-boundary pairs, and 65 other unmatched predictions. This grouping compares predicted boundaries and types with the reference annotations. A relaxed match still requires the same type and an IoU of at least 0.5.

Two unadapted Qwen outputs illustrate these distinctions. For 1801-s0004, the prediction is the S span 参与公司无人机飞控系统技术项目开发策划 (participate in planning a technical development project for the company's UAV flight-control system), whereas an overlapping reference S span is 技术项目开发策划 (technical project development planning). Their overlap is below the relaxed threshold, so this pair matches under neither criterion. For 1802-s0005, both layers select 方法 (method), but the prediction is S and the reference is K. This pair has identical boundaries but fails both typed matching rules. Appendix C gives the complete source sentences and offsets for these prediction-reference comparisons.

The exploratory expanded-pool Qwen runs produced different rankings across evaluation targets. With seed 42, the adopted expanded pool yielded F1 of 0.809 on its Silver test set and 0.585 on the shared human reference, whereas the alternative yielded 0.759 and 0.597, respectively (Table 18). The adopted pool had the higher Silver-test score, and the alternative had the higher human-reference score. The pool-specific Silver tests differ in both texts and labels. The annotation-quality analysis provides the common-sentence comparison against blinded human annotations. Table 19 gives the constituent batches and final counts.

Exploratory expanded JobBERT results in Appendix C show lower human-reference scores under jointly changed supervision, source composition, and model selection.

DISCUSSION

Chinese-SkillSpan makes two annotation decisions explicit: which source characters form a competency mention and which type the mention receives. This allows boundary and type errors to be examined alongside aggregate scores. The different results on silver and human targets show why agreement with generated labels cannot replace human-reference evaluation.

The matched Qwen experiments support a benefit from silver supervision relative to the same unadapted model under fixed inference conditions. The adapted Qwen mean remained below the observed score of the Silver-trained JobBERT-zh + CRF baseline. Prompted GPT retained the highest observed human-reference score. Training and inference settings differ across these model families.

Limitations

Reference design and label scope. The human reference set came from a model-disagreement queue and informed handbook development. It therefore supports diagnostic comparisons rather than a development-independent, representative evaluation of recruitment texts. The separate 50-sentence blinded study measures agreement among annotators applying the final handbook. All three annotators contributed to handbook development, so their agreement reflects an experienced team's use of the rules. K includes qualification and industry-background proxies, while L has only two reference spans. Broader evaluation requires independent annotation of texts from new source documents and separate human review of K subtypes.

Label versions and data exposure. The human reference set, historical supervision, and blinded agreement study use different handbook versions. Each set of results applies to the label version used in that experiment. The original corpus partitions were document-disjoint, and checks on the 2,150/169 Silver training/development sets excluded reference sentences by identifier or exact/NFC-normalized text. Later experimental splits nevertheless share source-document groups. These results therefore do not establish generalization to unseen source documents. Incomplete historical pretraining records also leave prior exposure to evaluation texts uncertain.

Experimental scope. The Qwen comparison jointly measures output validity and span extraction. Prompted configurations differ in inference budgets and access routes. The final expanded-pool comparisons use a single training seed, while the three-seed main experiments provide limited evidence about training variability. Source-group bootstrap intervals describe variation within this reference set and these model runs. False-positive analysis is limited to eight sentences with no reference spans.

Model assistance. Qwen, DeepSeek, and Kimi were evaluated as baselines and also supplied suggestions during assisted annotation. Shared instructions and machine-assisted review may carry common annotation preferences into the reviewed labels. Complete model-version records are unavailable for some suggestion tools. The blinded annotation study provides complementary evidence with machine suggestions hidden.

Future directions. Within the available annotation budget, this study focuses on flat, single-label spans with explicit source offsets. Future work should prioritize independent evaluation on new source documents and could explore overlapping or multiply labelled spans, together with job titles and document-level context. Applications requiring concise competency descriptions could be addressed through a separate normalization or summarization task linked to supporting source spans. Such extensions would require task-specific annotation and scoring protocols, while retaining the current extraction task as a separate evaluation track.

Responsible Use

The annotations describe requirements in job advertisements, not applicants' abilities. The reported extraction results do not validate the resource for assessing or ranking individual applicants. Qualification and industry-background labels in K describe employer requirements; treating them as measures of applicants' competence could reproduce credential-based or sector-based selection biases. Applicant screening would require a separate fairness and validity assessment. Appendix D documents the scope of automated privacy screening and the candidates requiring human review.

CONCLUSIONS

Chinese-SkillsSpan provides a benchmark for identifying and typing competency spans in Chinese job advertisements. It combines source-text annotations, separate human reference labels and silver supervision, and an ESCO-informed handbook for span boundaries and context-dependent type assignment. Character offsets, released predictions, and a shared scorer support inspection of boundary and type errors. A separate blinded study assessed independent annotation agreement under the final handbook, distinct from assisted label review.

After individual self-review, the three annotators achieved a mean pairwise typed exact-span F1 of 0.818. Under fixed inference conditions, Qwen adaptation improved end-to-end extraction relative to the same unadapted model, while the adapted mean remained below the Silver-trained JobBERT-zh + CRF result and prompted GPT had the highest observed typed exact-span micro-F1 on the human reference set. The encoder comparison did not establish an advantage for ESCO initialization under the evaluated protocol. Together, the resource and evaluation protocol support diagnosis of boundary and type errors under explicit annotation rules.

DATA AVAILABILITY

The dataset archive (v0.1.3; DOI: 10.5281/zenodo.22698504) contains the corpus, annotation handbook, 150-sentence human reference set, and Silver training and development files used by Qwen and the encoder comparison (2,150/169 records). The repository archive series has concept DOI 10.5281/zenodo.22288337; the version-specific link above identifies the dataset used here. A separate historical annotation-layer release provides human-annotation and additional-review exports with pseudonymized annotator identifiers, original texts, and offsets. The release for the independent blinded three-annotator study contains frozen individual annotation layers, source identifiers, agreement code, and results. These later annotation releases are maintained separately on GitHub.

The Qwen reproduction archive (DOI: 10.5281/zenodo.22851581) contains the three Qwen LoRA adapters trained on these Silver sets, selected expanded-silver checkpoints, predictions, configurations, and scoring tools. For the adopted 9,540-record expanded Silver pool and the supplementary 9,646-record alternative, public materials comprise split identifiers and selected model outputs. Complete texts and labels are retained locally and are available to editors and reviewers through the corresponding author for confidential peer review. Six XLM-R/ESCOXLM-R checkpoints trained with Chinese supervision are linked from the Hugging Face model index; their code, configurations, and predictions are provided in the audit archive (DOI: 10.5281/zenodo.22942441). The reproduction documentation distinguishes local rescoring of archived predictions from checkpoint-inference results reported by the server.

The JobBERT-zh initialization release provides the domain-adapted encoder and inherited CRF. The Silver-trained JobBERT-zh release provides the continued-training checkpoints, with seed 42 as the default. The project repository and reproduction guide link the code, preprocessing and scoring procedures, and experiment records. Table 21 summarizes component access and verification. External-task protocols and results are retained in the supplementary experiment archive. Reuse is governed by component-specific terms. Software and documentation explicitly covered by Apache-2.0 retain that licence. Author-controlled annotations and research materials may be used for academic research and peer review with attribution. Reuse of third-party recruitment text follows the applicable source terms and permissions. Model checkpoints and adapters remain

subject to their accompanying notices and base-model licences. The data-access guide specifies the applicable terms, and Appendix D records earlier versions and access limitations.

APPENDIX A. CODING GUIDE AND EXAMPLES

This guide follows Handbook v4.2.14 and complements Figure 4. The reader-facing v4.2.14 edition clarifies examples for interpretation of the current rules; unresolved cases retain that status. This documentation does not replace the studies’ frozen handbooks or labels. The source package contains the bilingual manuals and crosswalk; Section A.4 records version boundaries.

A.1 Label definitions and notation

Table 2 summarizes the four label definitions and current examples. In the appendix examples, square brackets mark selected source substrings, with a separate type code after each span. Brackets, colours, and type codes are display aids and are not source characters; boundaries remain identifiable in grayscale. Examples illustrate the handbook, not an additional evaluation sample. *Source key.* E: ESCO’s skills hierarchy and transversal concepts (European Commission, 2024); Q: EQF definitions (Council of the European Union, 2017); Z: related annotation guidance in SkillSpan, Appendix B (Zhang et al., 2022b); I: unitization and categorization in agreement analysis (Artstein and Poesio, 2008); P: our V4.2.14 operational decision. Combined marks indicate adaptation, not verbatim rules; Chinese examples and exact boundaries are project illustrations.

A.2 General annotation decisions

G1–G15 are quick-reference identifiers. Their source sections and amendment IDs (R08–R23) are mapped in the release; they do not replace the handbook’s rule numbering.

Table 11. Annotation principles for target identity and complete source boundaries.

Rule	Annotation principle	Chinese example
G1	The annotation scheme distinguishes L, K, S, and T. Collapsed categories are used only in secondary evaluation. ^[E,P]	[英语六级] [L] (College English Test Band 6); [计算机专业] [K] (computer science major); [数据分析] [S] (data analysis); [责任心] [T] (sense of responsibility)
G2	Final annotations contain non-overlapping spans. A sentence may contain several independent mentions, with each character assigned to at most one span. ^[P]	熟悉使用 [Word] [S], [Excel] [S], [PPT] [S] 等办公软件 (Familiar with using Word, Excel, PowerPoint, and other office software.) No additional covering list span.
G3	Each mention is a contiguous source substring that preserves spelling and internal spaces. Any normalization is stored separately. ^[Z,P]	熟悉 [Jekins] [S] (familiar with Jekins) Keep the source spelling, not Jenkins.

Table 11 (continued)

Rule	Annotation principle	Chinese example
G4	Annotators select the shortest span that expresses a complete requirement. Words and identifiers remain intact; completeness takes priority over length, with no fixed length cap. ^[Z,P]	[支持服务] [S] (support services) Not the fragment 支持服 (a truncation of “support services”).
G5	Generic proficiency triggers usually remain outside the span, while heads needed to express the requirement remain inside. Excluded triggers can still inform type assignment. ^[Z,P]	会使用 [Excel] [S] (able to use Excel); [会聊天] [T] (can converse); 有 [服务意识] [T] (has service awareness)
G6	Coordinated items are separated when each remains meaningful in context. Shared qualifiers may stay outside the spans, and implied words are not added. ^[Z,P]	熟悉使用 [Word] [S], [Excel] [S], [PPT] [S] 等办公软件 (Familiar with using Word, Excel, Power-Point, and other office software.) Shared-experience case: see A.3.
G7	A necessary action, object, or shared head remains within the span when splitting would change its meaning. Span length alone does not determine type. ^[Z,P]	[优化曝光与转化率] [S] (Optimize exposure and conversion rate.) Isolated metric names would lose the optimization activity.

Table 12. Annotation principles for scope, experience, type, and unresolved records.

Rule	Annotation principle	Chinese example
G8	Benefits, company promotion, administrative instructions, experience durations, and pure outcomes are outside competency scope on their own. Mixed clauses are assessed separately. ^[Z,P]	五险一金, 带薪年假 → []; (Five types of social insurance and a housing fund; paid annual leave.) [分析推广效果] [S], 保证流量/用户增长 (Analyze promotion effectiveness; ensure traffic/user growth.)
G9	Experience wording is omitted only when the remaining activity preserves the requirement. Necessary practical-experience wording is retained. Explicit industry background follows the handbook’s broad-K convention. Experience wording is never added to the source. ^[P]	[机器人竞赛经验] [S] (Robotics competition experience.) Retain the suffix if a bare competition name changes the requirement; do not label generic 经验 (experience) alone.
G10	Education qualifications include their threshold wording. Language examinations and certificates receive L; non-language qualifications receive K. ^[Z,P]	[本科及以上学历] [K] (bachelor’s degree or higher); [大学英语 6 级] [L] (College English Test Band 6); 了解 [ISO 27001] [K] (familiar with ISO 27001) Standard familiarity, not a claim of personal certification.

Table 12 (continued)

Rule	Annotation principle	Chinese example
G11	Tool and method mentions are typed from their predicate context. Occupational use supports S; principles or theory support a complete K phrase. Familiarity alone does not determine type. [Q,P]	使用 [SQL] [S] (use SQL); 了解 [SQL 原理] [K] (understand SQL principles)
G12	Occupational functions are distinguished from general cognition, communication, and coordination. A duty clause may contain T; usefulness across industries alone does not establish T. [E,P]	[编码能力] [S] (coding ability); [沟通能力] [T] (communication skills); [协调外部资源] [T] (coordinate external resources)
G13	Scope, boundaries, and type are resolved separately, without a total label priority. Once scope and boundaries are valid, type-only uncertainty may receive provisional S with alternatives logged; this is excluded from final training targets. [I,P]	Heading alone: 能力提升 (capability improvement) Insufficient scope evidence: record as unresolved, not a confirmed empty negative.
G14	Offsets count original Unicode code points, beginning at zero and excluding the end position. The indexed substring must reproduce the mention exactly; segmentation is a derived view. It does not replace the fixed human-reference offsets. [P]	text[start:end] == span Keep the original text, internal spaces, and offsets consistent.
G15	Alternatives and overlaps are recorded separately from the final flat layer. An unresolved record is excluded in full from training, with confirmed empty, missing, and unresolved annotations distinguished. [P]	五险一金 (five types of social insurance and a housing fund) → confirmed []; 能力提升 (capability improvement) without context → unresolved. Known partial spans do not make an unresolved record training-ready.

Interpreting coordinated capabilities. G6–G7 test whether each selected phrase expresses a requirement in context. They do not require every item to repeat an implied capability noun. For example, 协调 (*coordination*) can express a capability independently, whereas 曝光 (*exposure*) alone does not express the optimization action in 优化曝光与转化率 (*optimize exposure and conversion rate*). Figure 4 shows the recorded historical boundaries; it is not a relabeling under the final handbook. Any unresolved historical case requires a new human decision before its reference span is changed.

A.3 Shared experience: split only when meaning survives

The V4.2.14 amendment (R23) clarifies a shared-experience construction. [P] Each item remains identifiable in context; exclude the shared experience qualifier and do not add a covering span. The words for internet-company operations denote an occupational activity in this sentence; a bare requirement for experience in the internet industry would instead use the project K proxy rule. The presence of an industry name alone does not determine type.

4、三年以上大中型 [互联网公司经营] [S] 或 [BI] [S] 相关工作经验, [互联网公司广告运营分析] [S] 优先;

Three or more years of operations or BI experience in large or medium-sized internet companies; internet-company advertising operations analysis preferred.

Selected offsets: [9, 16), [17, 19), and [26, 37). BI is S in this context, not by a universal dictionary rule. In contrast, G7 retains the shared action in 优化曝光与转化率 (*optimize exposure and conversion rate*); splitting must not leave bare outcomes. The earlier long span in the initial reviewed LLM-annotation dataset remains historical rather than being silently overwritten.

A.4 Version boundaries

The human-reference metadata identify v4.2.10, while the original bulk-generation prompt follows v4.2.12. The current handbook is v4.2.14. Later amendments do not retrospectively change completed annotations. Inspection confirmed v4.2.10 metadata on all 150 reference records; the Silver training/development rows do not carry individual handbook-version fields. One documented difference is present in Silver training record 1838-s0008: its historical two-span annotation retains a shared experience phrase that R23 later splits into three spans. This named record is absent from the human reference set and the Silver development set. A rule-family screen covers all 2,469 reference, training, and development records, but flags candidates for contextual review rather than proving semantic compatibility. The original targets and scores remain unchanged. Figure 4 reproduces frozen v4.2.10 human-reference annotations, whereas Table 2 and Tables 11–12 illustrate decisions under the current v4.2.14 handbook. Later rule amendments do not retrospectively alter the historical annotations or reported scores.

The label-version documentation specifies the handbook and annotation files used in each experiment. The accompanying manuals preserve original and revised spans and their rationale. Human-reviewed original Silver remains supervision unless separately designated and isolated for evaluation. Evaluation uses the frozen reference, shared protocol, occurrence schema, parser, and scorer; the reproduction guide maps these terms to their released files (Appendix D).

APPENDIX B. ANNOTATION-QUALITY EVIDENCE

Historical coding and assisted-review comparisons below use designs distinct from the independent blinded study in Table 4. The initial 50-sentence calibration study involved independent raw-text coding before adjudication under the historical handbook. The sentences were sampled from the 980-record conflict queue. Available exports reproduce 98 exact matches between 211 and 230 spans, giving F1 0.444. Character-level κ is 0.567 for L/K/S/T plus O, over all 2,571 source code points, including spaces and punctuation. All 50 identifiers belong to the calibration portion of the human reference set and none occurs in the Silver training or development identifiers. This calibration study supplies independent-coding evidence for a subset of the human reference set. These calculations use the available later exports; exact correspondence with the original frozen files has not been verified. The agreement calculations provide the supporting counts. The reference's challenge cohort contributes 415 spans (290 S, 83 K, 41 T, one L), and the calibration cohort contributes 248 (158 S, 42 K, 47 T, one L). Seven challenge sentences and one calibration sentence have no annotated span. The identifiers contain 61 source-document prefixes, verified against the original corpus identifiers. The adjudicated reference remains separate from the original coding and assisted-review layers. A retrospective comparison of the two available

individual calibration exports with their adjudicated 50-sentence reference gives typed exact F1 of 0.640 and 0.606 (boundary F1 0.766 and 0.684). Because these annotators contributed to the reference, this comparison measures changes introduced by adjudication. Character-level Cohen's κ adjusts observed agreement for agreement expected by chance from the two coders' marginal label frequencies across the source character positions:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \quad p_e = \sum_{c \in \mathcal{C}} p_A(c) p_B(c). \quad (4)$$

Here, p_o is the fraction of source character positions with identical labels, and $p_A(c)$ and $p_B(c)$ are the two coders' label proportions. The label set $\mathcal{C}$ comprises L/K/S/T and O. For the four-label scheme plus O, the available exports yield $p_o = 0.771$ and $p_e = 0.471$, giving $\kappa = 0.567$. These coefficients use character positions rather than the spans counted in Eq. (2). The 2,571 positions are nested within the 50 sampled sentences.

Empty-empty sentences contribute no span true positives. Span matching and character-level agreement use different units; reproduction materials document optional category projections. The post-generation review compares machine-prefilled annotations with the human-reviewed layer: 140 exact typed matches among 144 machine and 148 reviewed spans give P/R/F1 of 0.972/0.946/0.959. Ninety-five sentences have identical complete span sets. The sample was stratified by historical source, empty/nonempty status, and text length from 2,179 eligible records. The frame excludes 100 initially reviewed records, 100 records reserved for an earlier pilot, and 72 pending records from the 2,451-record original Silver pool; two small strata totalling 11 records were unsampled. The estimate is conditional on this stratified review sample.

The 15-sentence assisted-review subset is nested within the post-generation Silver review cohort. Both annotators had access to machine coding suggestions. Their human layers contain the same 27 spans, yielding exact span F1 and character kappa of 1.000. The shared suggestions may contribute to this agreement between assisted reviewers.

Input and Span Distributions

Median input lengths are 33 Unicode code points in each Silver training/development set used by Qwen and the encoder comparison, 41.5 in the reference, and 33.5 in the additional review sample. Median span length is six code points in supervision and both review layers, compared with four in the reference. Counts include spaces and punctuation; Figure 3(a,b) shows these historical distributions. Input-length distributions include records with no annotated spans. Span-length densities are estimated from all observed spans in each layer and scaled to equal maximum widths. The detailed view covers 0–20 Unicode code points, containing 96.4–100.0% of spans in each layer; the overview retains the full range. White dots mark medians and thick lines mark interquartile ranges.

Table 13 reports counts for the full adopted expanded pool and its Qwen partitions. Counts use the experimental files, including repeated records.

Table 13. Competency-span counts in the adopted expanded Silver pool and its Qwen splits.

Split	Records	No spans	L	S	K	T	Total spans
Training	8,842	2,321 (26.2%)	149	10,653	4,887	3,851	19,540
Validation	348	91 (26.1%)	2	403	184	152	741
Test	350	80 (22.9%)	4	472	194	150	820
Full pool	9,540	2,492 (26.1%)	155	11,528	5,265	4,153	21,101

No spans gives the number and percentage of records with empty annotations.

Additional Review of Newly Generated Silver Labels

Two reviewers submitted separate annotation layers for the 150 preselected sentences. Their normalized texts have no overlap with the 350 sentences in the human reference and initial and post-generation reviews. The sample includes 60 cloud, 45 public-sector, and 45 listed-company sentences. Both reviewers could see suggestions from the Codex and Cursor configurations in Table 5, so their agreement reflects assisted review.

Table 14 summarizes the competency types in the two review layers. Table 15 compares the pre-adjudication span sets. Its percentile 95% intervals use 10,000 sentence resamples within source strata, conditional on the source quotas. The calculation notes give the random seed and export checks.

Table 14. Competency composition in the additional review sample: span counts (within-reviewer percentages).

Reviewer	L	S	K	T	Total spans
A	3 (0.8%)	217 (60.6%)	77 (21.5%)	61 (17.0%)	358
B	3 (0.8%)	218 (59.7%)	81 (22.2%)	63 (17.3%)	365

Table 15. Agreement in the additional review sample.

Comparison	Exact F1	95% interval	Identical sentences
Reviewer A–Reviewer B	0.816	[0.769, 0.860]	101/150
Codex–Cursor suggestions	0.962	[0.937, 0.983]	138/150
Reviewer A–Codex suggestions	0.823	[0.776, 0.866]	100/150
Reviewer B–Codex suggestions	0.850	[0.803, 0.892]	108/150

Agreement is computed from both submitted layers, including the 49 sentences with differing span sets. Character-level calculations assign O/L/K/S/T to original Unicode-code-point positions, including spaces and punctuation. Two positions have conflicting types within one reviewer layer and are excluded from both character sequences, leaving $N = 5,990$ positions. All original spans remain in span-level F1. The accompanying agreement data identify the affected record. Later adjudication and incorporation into training targets have not been verified. Observed agreement over the remaining character positions is $p_o = 0.934$, with $p_e = 0.439$ and $\kappa = 0.883$.

For nominal Krippendorff's α , let m_c count assignments to category c across both reviewers, with $\sum_c m_c = 2N$. The observed and expected disagreements are

$$\alpha = 1 - \frac{D_o}{D_e}, \quad D_o = 1 - p_o, \quad D_e = 1 - \frac{\sum_c m_c(m_c - 1)}{2N(2N - 1)}. \quad (5)$$

Here $D_o = 0.066$ and $D_e = 0.560$, giving $\alpha = 0.883$. The character coefficients and span F1 describe different units of agreement (Artstein and Poesio, 2008); the many O positions contribute to the character calculation. The machine-machine comparison measures concordance between the suggestion layers. Export hashes and calculation code are available with the reproduction materials.

Blinded Coding by Three Annotators

The annotators comprised one human resource management specialist with a doctorate and two employed master's students in management. The students had sustained annotation experience and three to five years of professional experience. All three contributed to handbook development; two professors advised on the coding process and handbook preparation.

The formal sample contains eight listed-company, 16 public-institution, and 26 Tianchi sentences. The research team reports random selection, with one sentence per advertisement identifier. Each annotator completed familiarization and training before independent coding. The released annotations identify the sampled sentences and sources. The candidate frame and sampling log are unavailable, so the original draw cannot be reproduced. Source-stratified bootstrap intervals quantify agreement uncertainty within this sample. Five- and 15-sentence training sets are excluded from formal agreement. Annotators then individually reviewed their own labels using the handbook, with machine suggestions and other annotators' labels hidden throughout. Mean pairwise typed exact-span F1 was 0.663 before self-review and 0.818 afterward; the analysis uses the latter labels. Both stages are available in the reproduction notes. The historical model reference retains its original labels.

The three coders in Table 4 produced 104/123/113 spans; exact typed matches numbered 90/92/96 for A-B/A-C/B-C. Pairwise F1 ranges are 0.762–0.805 for K, 0.778–0.869 for S, and 0.865–1.000 for T. All coders identified the same two L spans in one sentence. O accounts for 63.2–64.1% of character positions. The reproduction materials include individual annotations, per-type counts, and bootstrap code.

Silver Labels against Blinded Annotations

We compared the final expanded-pool labels with each of the three blinded annotation layers on the same 49 retained sentences (Table 16). One of the 50 study sentences remained unresolved at the candidate stage and was absent from both final pools; it was excluded from this comparison rather than scored as an empty annotation. Each per-annotator score pools exact boundary-and-type matches across the 49 sentences, and the mean averages the three micro-F1 scores.

Table 16. Agreement between final expanded Silver labels and the blinded annotation layers.

Silver pool	Sentences	Annotator A	Annotator B	Annotator C	Mean
Adopted expanded pool	49	0.724	0.681	0.721	0.709
Alternative expanded pool	49	0.723	0.716	0.721	0.720

Typed exact-span micro-F1. Each column uses a separate human annotation layer on the same sentences; no consensus labels were constructed.

This post hoc comparison provides suggestion-blinded, sample-level evidence alongside the assisted reviews. The historical Silver labels and final-handbook human annotations use different label versions. The selected sample and participating annotators therefore define the scope of these

agreement scores; they do not estimate the accuracy of the entire Silver pool. The reproduction materials provide counts, input hashes, and code. The original 150-sentence model scores are unchanged.

APPENDIX C. SUPPLEMENTARY EVALUATION

This appendix gives the shared evaluation protocol, uncertainty and error analyses, the JobBERT supervision comparison, and exploratory expanded-pool results. The supporting records provide per-run results, scorer details, and earlier protocol comparisons. The earlier protocols differ in output format and demonstration settings.

Shared-Guideline Evaluation against the Human Reference

The shared protocol evaluates six configurations on all 150 reference sentences, using V4.2.12-based general instructions and five schematic examples. The system prompt, examples, and checksum manifest are included in the Qwen reproduction archive.

Table 5 lists the model configurations, access routes, and output budgets. For Claude Sonnet 4.5, the gateway request used `claude-sonnet-4-5`, and the response reported `claude-sonnet-4-5-20250929`. Request dates, response identifiers, and completion metadata are available in the reproduction notes.

Models return an ID and spans as literal text, zero-based occurrence, and L/K/S/T type. The span parser resolves spans to Unicode offsets for exact and relaxed scoring with the released span scorer. All input texts match the human reference set.

Qwen adaptation is evaluated with fixed inference settings, as specified in the main implementation section. The API configurations share task instructions but use the output budgets listed in Table 5.

Span parsing and matching. The parser resolves literal spans to source offsets, sorts candidates by start offset and original output order, and keeps the first compatible candidate. Later duplicates, same-boundary type conflicts, and overlaps are rejected. Partial responses retain accepted spans; fully rejected or malformed responses become empty predictions. All supplied sentences are included in scoring.

Exact matching deduplicates identical span tuples within each sentence. Relaxed matching traverses reference spans in source order and selects the unused same-type prediction with highest IoU, taking the first prediction on a tie. This greedy matching uses an IoU threshold of 0.5. Empty-empty records contribute no span counts. The public span scorer implementation and human-reference evaluation entry point document the executable scoring procedure. The completion and parser audit documents output-status counts. In the DeepSeek thinking run, 18 requests exhausted the output budget before producing a final answer. These responses are scored as empty predictions, with all 150 reference sentences retained. The resulting score includes completion failure under that budget.

Paired bootstrap intervals. To allow dependence among sentences with the same source identifier, we grouped the 150 reference sentences by the prefix preceding `-s`, giving 61 groups. Each of 10,000 paired bootstrap draws samples 61 complete groups with replacement and includes all their sentences. The same draw is used for all compared models and each fixed training seed; micro-F1 is recomputed before seed means and paired differences are calculated. Table 8 reports

percentile intervals. These prefixes match the source-document identifiers in the original corpus. The intervals account for within-document dependence and are conditional on the reference set and fitted models. They are unadjusted for multiple comparisons. Training-seed SD separately measures variation across training runs. As a sensitivity check, widening the bootstrap intervals for the six displayed comparisons using Bonferroni tail probabilities leaves all inclusion/exclusion-of-zero conclusions unchanged; detailed intervals and code are in the supplementary calculations.

Mismatch examples. The two examples in the Results section use Qwen predictions without an adapter under the shared prompt. For 1801-s0004, the complete sentence is (4) 负责参与公司无人机飞控系统技术项目开发策划、总体方案论证、技术设计 (responsible for participating in technical project development planning, overall design assessment, and technical design for the company's unmanned aerial vehicle flight-control system). The illustrated prediction is S at [5, 24) and the overlapping reference is S at [16, 24); their IoU is $8/19 < 0.5$.

For 1802-s0005, the complete sentence is 6、培训其他方法、工具和技术支持多个应用。(approximate gloss: training in other methods, tools, and techniques to support multiple applications). Both layers select 方法 (method) at [6, 8), with predicted type S and reference type K. Offsets are zero-based, half-open Unicode-code-point intervals. The second response also contains two rejected occurrence references. Both examples use the existing reference annotations.

Per-seed Qwen results. The per-seed record retains precision, recall, exact/relaxed F1, and calibration/challenge scores underlying Tables 9 and 8. Boundary-only exact F1 in Table 9 uses the same accepted spans, with types collapsed and duplicate coordinates removed within sentences. The model-diagnostic records provide per-seed and per-type precision, recall, TP/FP/FN, and the full-precision summaries used in Table 10; Figure 5 retains individual-seed Qwen results.

Qwen Prediction Diagnostics

Figure 5 reuses the same 150 reference sentences and the accepted spans from each run. Typed exact matching is completed on each full sentence before aggregation by reference-span length. Bins (≤ 4 , $5-8$, ≥ 9 Unicode code points) are fixed in advance; they contain 346/243/74 reference spans in 102/110/56 sentences, respectively. A sentence may contribute reference spans to multiple bins. Type-specific unmatched predictions count as FP even in empty-reference sentences. Results for each training seed are evaluated separately on the same reference sentences. On the eight empty-reference sentences, accepted false-positive spans decrease from 19 without an adapter to 6, 4, and 0 for seeds 42, 43, and 44; affected sentences number 2, 2, 1, and 0. L has two reference spans: the no-adapter run recovers 0/2 and each LoRA run 2/2, with one extra L prediction in every run. Complete TP/FP/FN, legal-empty-output counts, and parser-rejection diagnostics are available in the diagnostic records.

Historical JobBERT Label-Version Comparison

Supervision and model selection. The two JobBERT conditions start from the same released encoder and inherited CRF. Earlier supervision (archived as B1) uses historical extraction labels, whereas revised supervision (archived as B2) incorporates updated guidelines, LLM annotations, and adjudication. The conditions use the same training and development texts, with label changes in 1,440 training records and 107 development records. Checkpoints are selected by typed exact-span F1 against the corresponding development labels. The comparison therefore changes both training supervision and checkpoint-selection targets. Both conditions use the JobBERT training

846 settings in Table 6. The revised-label condition is the Silver-trained JobBERT-zh + CRF baseline
847 reported in the main results. The archived revised-label files for Qwen and the encoder compari-
848 son use 2,150/169 records, after the exclusions described in the main evaluation protocol; these
849 are distinct from the 2,156/169 JobBERT inputs and the 2,451-record original Silver candidate
850 pool.

Table 17. Historical JobBERT comparison of training supervision and checkpoint-selection labels on matched texts.

Supervision and selection labels	Train / dev records	Typed exact
Earlier labels	2,156 / 169	0.142 ± 0.014
Revised labels	2,156 / 169	0.554 ± 0.005

Human-reference F1 scores (0–1) are mean ± sample SD over seeds 42, 43, and 44. Both training and development labels change between conditions.

851 The historical JobBERT configuration scored higher with revised labels: mean typed exact-span
852 micro-F1 increased from 0.142 ± 0.014 with earlier labels to 0.554 ± 0.005 with revised labels
853 (Table 17). Both conditions used the same initial encoder and CRF and the same training and
854 development texts, but both label sets changed. The difference therefore reflects the combined
855 changes in training supervision and checkpoint-selection targets.

856 **Exploratory Expanded-Silver Experiments**

857 The expanded Qwen runs use seed 42 and the shared prompt, occurrence output, parser, and
858 Qwen training settings in Table 6. Loss applies only to assistant output. Both Qwen checkpoints
859 are selected at epoch one using Silver-validation exact F1; decoding is deterministic with a 2,048-
860 token output cap. The complete configurations retain the parameter records. The expanded Job-
861 BERT runs retain the domain-adapted encoder and inherited CRF. The adopted-pool JobBERT
862 run uses 8,842 training and 698 development records. Qwen instead uses 348 validation and 350
863 test records with the same 8,842-record training set.

864 Table 18 reports the two Qwen runs on their Silver validation and test sets and the human refer-
865 ence set.

Table 18. Expanded-pool Qwen results (seed 42). Silver columns measure agreement with generated targets; human-reference columns use human-finalized labels. Scores are typed exact micro-F1 except the final relaxed-F1 column.

Pool variant	Silver validation agreement	Silver test agreement	Human reference exact	Human reference relaxed
Alternative: 8,943 / 351 / 352	0.783	0.759	0.597	0.713
Adopted: 8,842 / 348 / 350	0.833	0.809	0.585	0.703

Counts give training / validation / test records. The adopted and alternative pools contain 9,540 and 9,646 records, respectively. Both combine earlier Silver supervision and public-prompt annotations; they differ in the annotation channel used for a 2,500-candidate batch and the records retained after checking. Checkpoints were selected at epoch one with patience two.

866 The adopted pool gives higher agreement with its Silver test labels and approximately 0.011 lower
867 exact F1 on the human reference set than the alternative pool. The replaced batch contains 2,353

accepted Codex annotations and 2,459 accepted API-gateway annotations, respectively. The score difference therefore combines changes in annotations and record inclusion. On the human reference set, expanded JobBERT obtains exact F1 of 0.242 with the alternative pool and 0.232 with the adopted pool (seed 42); the latter has relaxed F1 of 0.485. A separate configuration trained on 7,117 newly annotated records gives 0.237 ± 0.002 across three seeds. All are below the original Silver-trained JobBERT-zh + CRF result. Alongside training size, these experiments change the annotation protocol, source composition, and development set. For comparison, Qwen trained on the original 2,150-record Silver set scores 0.500 at seed 42 and averages 0.540 across three seeds. Expanded-pool Qwen therefore scores higher in these runs, whereas expanded JobBERT scores lower. Because supervision and checkpoint-selection targets change together, the results support an exploratory comparison of complete configurations. Further controlled experiments are needed to explain the JobBERT decline, including possible effects of label versions, input windowing, and decoding. The supporting experiment records include exploratory single-seed supervision-subset runs and their differing checkpoint-selection arrangements.

APPENDIX D. DATA LINEAGE, ACCESS, AND REPRODUCTION

Data Availability provides the release locations and reuse terms; Table 21 distinguishes the reproduction tasks supported by each component. The supplementary audit package supplies the calculations below, input hashes, and executable checks. The linked supplementary materials also contain earlier experiment outputs and prompts.

In released files and archived summaries, the human reference set and the additional review sample are identified as Gold150 and QA150, respectively. The original Silver pool is named Silver-plus in archived files. The Silver supervision labels used by the reported baselines correspond to the revised-label version mapped in Appendix C. The original Silver pool contains candidate records from which model-specific training/development sets were formed. The legacy identifiers provide a crosswalk to the descriptive names used in this paper.

Source Categories and Data Preparation

The four original collection labels describe acquisition batches; the three acquisition routes identify MacroData, public-institution collection, and Tianchi. Acquisition batches and routes classify the same material at different levels. The original training partition contains 10,312 graduate-recruitment and 7,148 AI sentences; development contains 2,143 AI sentences; test contains 1,423 AI, 473 Tianchi/cloud, and 1,341 public-institution sentences. The later expansion uses Tianchi/cloud, public-institution, and listed-company source groups. The blinded agreement sample's 26/16/8 counts use these later source groups.

MacroData's operator is Chongqing Maherui Information Technology Co., Ltd. Public-institution records retain source names, titles, URLs, listed publication dates, and geographic fields where available. Publication dates are not collection dates; a complete crawl-window and source-exclusion history has not been recovered. The authors confirm academic-research use permission. Repository notices separately identify the rights in author-created annotations and the restrictions on third-party text.

Table 19 records the candidate and retained counts for the expanded pools. The preparation memo

preserves intermediate normalization counts and assembly reconciliation; the reported experiments use the final splits.

Table 19. Candidate and retained record counts in the expanded pools.

Component	Candidates	Adopted retained	Alternative retained
Earlier Silver supervision	Stage-specific	2,077	2,077
First two new batches	3,620	3,407	3,407
Third new batch	1,816	1,703	1,703
Channel-comparison batch	2,500	2,353	2,459
Final pool	—	9,540	9,646

New-candidate selection used requirement-related keywords, fixed hash ranking, and source quotas. HTML/whitespace cleaning, sentence splitting, and length filters preceded annotation. Exclusion checks used the stored sentence strings and available pretraining dump. The 3.2-million-line dump’s content hash was not retained, so the historical pretraining exposure cannot be fully reconstructed. Labels retain the handbook version used when they were generated.

Split Membership and Document Overlap

Original records use source-document identifiers, while expansion records use source-file and source-row identifiers. Overlap checks operate within these namespaces because no crosswalk between them is available.

Table 20. Shared source-document groups in the later experimental splits. The Silver train/development rows use the 2,150/169 split for Qwen and the encoder comparison. Counts refer to different sentences within shared groups, not identical sentences across splits.

Comparison	Shared groups	Left sentences	Right sentences
Silver train / development	1	4	27
Silver train / human reference	59	531	143
Silver development / human reference	0	0	0
Adopted expanded train / validation	185	1,060	208
Adopted expanded train / test	188	1,076	210
Alternative expanded train / validation	186	1,069	210
Alternative expanded train / test	189	1,079	211

No exact or NFC-normalized sentence is shared between the later Silver training/development sets used by Qwen and the encoder comparison or between training and the human reference set. NFKC normalization with whitespace removal and case folding finds two Silver training/development matches and one training/validation match in each expanded variant. No normalized training/test sentence matches were found in the expanded variants. The shared source-document groups are listed in Table 20.

The original 22,840-sentence partition is document-disjoint. The later experiments use sentence-level splits with the source-document overlap described above.

The Silver training set used by Qwen and the encoder comparison contains 1,910 distinct complete texts among 2,150 records. Repeated records remain weighted in training. A deduplicated

training comparison and document-disjoint evaluation would be needed to measure the effects of repeated records and source-document overlap.

Blinded-sample overlap. The 50-sentence blinded agreement sample has no ID or normalized complete-text match with the original Silver training/development sets (2,150/169) or the human reference set. Each expanded pool contains 46 of these sentences in training, two in validation, and one in testing; one is absent. Thus the independent annotations can support agreement analysis, but these sentences have already been seen by the expanded models. The membership audit records the frozen-input hashes and counts.

Resource Access and Privacy Screening

Table 21. Available components and supported reproduction tasks.

Component	Access route / available objects	Supported use and access conditions
Dataset archive v0.1.3	Texts and labels: source corpus, human reference (150 sentences), and Qwen/encoder Silver train/dev (2,150/169 records)	Download the named version to inspect the released labels and use the specified splits.
Qwen reproduction archive	Model archive: three Qwen LoRA adapters trained on the 2,150/169 Silver sets, selected expanded adapters, frozen predictions, prompts and scorer	Follow its model index for inference or rescoring; obtain base weights and supervision files from their respective sources.
Expanded pools	9,540 adopted / 9,646 supplementary records; complete files verified locally against split manifests	Published IDs and selected outputs support inspection and rescoring. Complete inputs are available through the corresponding author for confidential peer review.
Additional review and historical coding	Pseudonymized texts and human layers linked in Data Availability; source hashes and cohort membership	Recalculate agreement for the documented historical annotation and assisted-review procedures.
Three-annotator blinded study	Agreement release: 50 sentences, three individual label layers, source IDs, scorer and intervals	Recalculate agreement among individual annotations on the 50 formal sentences; the five familiarization and 15 practice sentences are documented separately.
Encoders with Chinese supervision	Six public Hugging Face checkpoints; audit DOI 10.5281/zenodo.22942441	Frozen predictions, code and configurations support rescoring. Local rescoring verified all six runs; server checkpoint re-inference is separately recorded.

Data Availability states the component-specific reuse terms. Reproduction scope follows the released materials and access conditions listed above.

An automated screen examined text fields in 15 public corpus and annotation files, covering 28,635 distinct complete texts. Email, mobile-number, and landline patterns flagged 38, 74, and 50 texts, respectively; categories overlap. No such candidates were found in the human reference set or the additional review sample, but candidates occur in the Silver training/development sets and in the separate blinded agreement sample. These are pattern matches requiring human review, including possible organizational contact details or false positives. The screen covers the specified contact-information patterns; broader personal-information and redistribution-permission checks require separate review. The public audit reports aggregate counts; the masked record-level review list is retained locally for resolution before a privacy-vetted release.

Model reconstruction. JobBERT-zh is the project’s released Chinese recruitment-domain encoder, with an inherited CRF used in the reported continuation runs. The exact original tokenizer revision, some pretraining/selection settings, and truncation or windowing counts were not recovered. Its 256-token limit is not a character limit; the reference retains all target spans. The original Qwen Hub commit was not recorded; archived file hashes identify the supplied artifacts. These gaps limit training reconstruction; released predictions can still be rescored.

ACKNOWLEDGMENTS

OpenAI Codex, Claude, and Grok assisted with manuscript preparation and code, as detailed in Use of AI Tools. The authors reviewed the revisions and take full responsibility for the final manuscript.

FUNDING

This research was supported by the National Social Science Fund of China under the project “Research on the Interaction Mechanism between Enterprise Employees’ Social Capital and Human Capital in the Intelligent Era and Its Impact on Career Success” (Grant No. 21BGL142). The grant supported API usage, research labor, and research meetings and access arrangements. The funder had no role in designing the study, collecting and analysing the data, preparing the manuscript, or deciding whether to submit the article for publication.

COMPETING INTERESTS

The authors declare that they have no competing interests.

REFERENCES

- Artstein, R. and Poesio, M. (2008). Inter-coder agreement for computational linguistics. *Computational Linguistics*, 34(4):555–596.
- Bender, E. M. and Friedman, B. (2018). Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6:587–604.
- Bhola, A., Halder, K., Prasad, A., and Kan, M.-Y. (2020). Retrieving skills from job descriptions: A language model based extreme multi-label classification framework. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 5832–5842. International Committee on Computational Linguistics.

- Cao, L., Zhang, J., Ge, X., and Chen, J. (2021). Occupational profiling driven by online job advertisements: Taking the data analysis and processing engineering technicians as an example. *PLOS ONE*, 16(6):e0253308.
- Chen, C. and Kong, F. (2021). Enhancing entity boundary detection for better Chinese named entity recognition. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, pages 20–25. Association for Computational Linguistics.
- Chen, X., Qin, C., Fang, C., Wang, C., Zhu, C., Zhuang, F., Zhu, H., and Xiong, H. (2024). Job-SDF: A multi-granularity dataset for job skill demand forecasting and benchmarking. In *Advances in Neural Information Processing Systems*, volume 37, pages 129329–129356. Curran Associates, Inc. Datasets and Benchmarks Track.
- Conneau, A., Khandelwal, K., Goyal, N., Chaudhary, V., Wenzek, G., Guzmán, F., Grave, E., Ott, M., Zettlemoyer, L., and Stoyanov, V. (2020). Unsupervised cross-lingual representation learning at scale. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8440–8451. Association for Computational Linguistics.
- Council of the European Union (2017). Council Recommendation of 22 May 2017 on the European Qualifications Framework for lifelong learning and repealing the recommendation of the European Parliament and of the Council of 23 April 2008 on the establishment of the European Qualifications Framework for lifelong learning. Official Journal of the European Union, C 189, pp. 15–28. Accessed: 2026-08-28.
- De Santo, A., Malandri, L., Mercorio, F., Mezzanzanica, M., and Nobani, N. (2026). SkillLens: Recognising and mapping novel skills from millions of job ads across Europe using language models. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 5: Industry Track)*, pages 877–885. Association for Computational Linguistics.
- Devlin, J., Chang, M.-W., Lee, K., and Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186. Association for Computational Linguistics.
- Ding, B., Qin, C., Liu, L., Chia, Y. K., Li, B., Joty, S., and Bing, L. (2023). Is GPT-3 a good data annotator? In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 11173–11195. Association for Computational Linguistics.
- European Commission (2024). ESCO skills pillar: Skills and competences. https://esco.ec.europa.eu/en/classification/skill_main. Version 1.2.0 used to define the annotation inventory; accessed: 2026-08-28.
- Fetahu, B., Kar, S., Chen, Z., Rokhlenko, O., and Malmasi, S. (2023). SemEval-2023 task 2: Fine-grained multilingual named entity recognition (MultiCoNER 2). In *Proceedings of the 17th International Workshop on Semantic Evaluation (SemEval-2023)*, pages 2247–2265. Association for Computational Linguistics.
- Geburu, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., and Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, 64(12):86–92.
- Gnehm, A.-S., Bühlmann, E., Buchs, H., and Clematide, S. (2022). Fine-grained extraction and classification of skill requirements in German-Speaking job ads. In *Proceedings of the Fifth*

- Workshop on Natural Language Processing and Computational Social Science (NLP+CSS), pages 14–24. Association for Computational Linguistics.
- Gururangan, S., Marasović, A., Swayamdipta, S., Lo, K., Beltagy, I., Downey, D., and Smith, N. A. (2020). Don't stop pretraining: Adapt language models to domains and tasks. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 8342–8360. Association for Computational Linguistics.
- Hripcsak, G. and Rothschild, A. S. (2005). Agreement, the F-measure, and reliability in information retrieval. *Journal of the American Medical Informatics Association*, 12(3):296–298.
- Hu, E. J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., and Chen, W. (2022). LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*.
- Kim, K. W., Kim, J.-D., and Aizawa, A. (2026). Refining and reusing annotation guidelines for LLM annotation. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 37951–37964. Association for Computational Linguistics.
- Klie, J.-C., Eckart de Castilho, R., and Gurevych, I. (2024). Analyzing dataset annotation quality management in the wild. *Computational Linguistics*, 50(3):817–866.
- Lafferty, J., McCallum, A., and Pereira, F. C. N. (2001). Conditional random fields: Probabilistic models for segmenting and labeling sequence data. In *Proceedings of the Eighteenth International Conference on Machine Learning*, pages 282–289. Morgan Kaufmann.
- le Vrang, M., Papantoniou, A., Pauwels, E., Fannes, P., Vandensteen, D., and De Smedt, J. (2014). ESCO: Boosting job matching in Europe with semantic interoperability. *Computer*, 47(10):57–64.
- Li, M., Shi, T., Ziemis, C., Kan, M.-Y., Chen, N., Liu, Z., and Yang, D. (2023). CoAnnotating: Uncertainty-guided work allocation between human and large language models for data annotation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 1487–1505. Association for Computational Linguistics.
- Lu, Y., Liu, Q., Dai, D., Xiao, X., Lin, H., Han, X., Sun, L., and Wu, H. (2022). Unified structure generation for universal information extraction. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5755–5772. Association for Computational Linguistics.
- Luo, H., Jin, Y., Wang, Y., Li, X., Shang, T., Liu, X., Chen, R., Wang, K., Salam, H., Wen, Q., and Liu, Z. (2025). DynamicNER: A dynamic, multilingual, and fine-grained dataset for LLM-based named entity recognition. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 16511–16535. Association for Computational Linguistics.
- MacroData (n.d.). MacroData. <https://www.macrodatas.cn/>. Platform operated by Chongqing Maherui Information Technology Co., Ltd. Accessed 20 September 2026.
- Magron, A., Dai, A., Zhang, M., Montariol, S., and Bosselut, A. (2024). JobScape: A framework for generating synthetic job postings to enhance skill matching. In *Proceedings of the First Workshop on Natural Language Processing for Human Resources (NLP4HR 2024)*, pages 43–58. Association for Computational Linguistics.
- Nguyen, K., Zhang, M., Montariol, S., and Bosselut, A. (2024). Rethinking skill extraction in the job market domain using large language models. In *Proceedings of the First Workshop on Natural Language Processing for Human Resources (NLP4HR 2024)*, pages 27–42. Association for

Computational Linguistics.

Peng, N. and Dredze, M. (2015). Named entity recognition for Chinese social media with jointly trained embeddings. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 548–554. Association for Computational Linguistics.

Podjanin, A. and Strietska-Ilina, O., editors (2020). *The feasibility of using big data in anticipating and matching skills needs*. International Labour Office, Geneva. Contributions from the ILO workshop held on 19–20 September 2019. Accessed: 2025-10-28.

Qwen Team (2025). Qwen2.5 technical report. arXiv preprint arXiv:2412.15115.

Sainz, O., García-Ferrero, I., Agerri, R., Lopez de Lacalle, O., Rigau, G., and Agirre, E. (2024). GoLLIE: Annotation guidelines improve zero-shot information-extraction. In *The Twelfth International Conference on Learning Representations*, volume 2024, pages 47083–47107.

Sayfullina, L., Malmi, E., and Kannala, J. (2018). Learning representations for soft skill matching. In *Analysis of Images, Social Networks and Texts*, volume 11179 of *Lecture Notes in Computer Science*, pages 141–152. Springer.

Senger, E., Zhang, M., van der Goot, R., and Plank, B. (2024). Deep learning-based computational job market analysis: A survey on skill extraction and classification from job postings. In *Proceedings of the Workshop on Natural Language Processing for Human Resources (NLP4HR)*, pages 1–15. Association for Computational Linguistics.

Tianchi (n.d.). Recruitment Dataset. <https://tianchi.aliyun.com/dataset/163746/>. Title translated from Chinese; accessed 20 September 2026.

Ul Haq, M. U., Rigoni, D., and Sperduti, A. (2026). A systematic comparison of large language models for data annotation in NER tasks. In *Proceedings of the Fifteenth Language Resources and Evaluation Conference*, pages 524–548. ELRA Language Resource Association.

Wang, S., Sun, X., Li, X., Ouyang, R., Wu, F., Zhang, T., Li, J., Wang, G., and Guo, C. (2025). GPT-NER: Named entity recognition via large language models. In *Findings of the Association for Computational Linguistics: NAACL 2025*, pages 4257–4275. Association for Computational Linguistics.

Xu, L., Yu tong, Dong, Q., Liao, Y., Yu, C., Tian, Y., Liu, W., Li, L., Liu, C., and Zhang, X. (2020). CLUENER2020: Fine-grained named entity recognition dataset and benchmark for Chinese. arXiv preprint arXiv:2001.04351.

Zaratiana, U., Tomeh, N., Holat, P., and Charnois, T. (2024). GLiNER: Generalist model for named entity recognition using bidirectional transformer. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5364–5376. Association for Computational Linguistics.

Zhang, M., Jensen, K. N., and Plank, B. (2022a). Kompetencer: Fine-grained skill classification in Danish job postings via distant supervision and transfer learning. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 436–447. European Language Resources Association.

Zhang, M., Jensen, K. N., Sonniks, S. D., and Plank, B. (2022b). SkillSpan: Hard and soft skill extraction from English job postings. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4962–4984. Association for Computational Linguistics.

Zhang, M., Jensen, K. N., van der Goot, R., and Plank, B. (2022c). Skill extraction from job postings using weak supervision. In *Proceedings of the 2nd Workshop on Recommender Systems for*

1112 *Human Resources*, volume 3218 of *CEUR Workshop Proceedings*.
1113 Zhang, M., van der Goot, R., Kan, M.-Y., and Plank, B. (2024). NNOSE: Nearest neighbor occu-
1114 pational skill extraction. In *Proceedings of the 18th Conference of the European Chapter of the*
1115 *Association for Computational Linguistics (Volume 1: Long Papers)*, pages 589–608. Associa-
1116 tion for Computational Linguistics.
1117 Zhang, M., van der Goot, R., and Plank, B. (2023). ESCOXML-R: Multilingual taxonomy-driven
1118 pre-training for the job market domain. In *Proceedings of the 61st Annual Meeting of the Asso-*
1119 *ciation for Computational Linguistics (Volume 1: Long Papers)*, pages 11871–11890. Associa-
1120 tion for Computational Linguistics.
1121 Zhou, W., Zhang, S., Gu, Y., Chen, M., and Poon, H. (2024). UniversalNER: Targeted distillation
1122 from large language models for open named entity recognition. In *The Twelfth International*
1123 *Conference on Learning Representations*, volume 2024, pages 12276–12294.